



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Project Task 6A

Network Design for Faculty of Computing Block N28B

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6.1 Introduction

Here comes the final step of the network design project for the Faculty of Computing, where we review its aims, scope, and objectives in this task. The primary goal is to develop an optimal floor plan that integrates a variety of networking devices and modern technologies, fully supporting education aligned with 4IR (Fourth Industrial Revolution) standards. The project is designed to create a cutting-edge and cost-effective network infrastructure that supports growth, ensuring students and staff have access to advanced technology in a comfortable environment for learning and collaboration.

Task 6 focuses on consolidating all the work completed in the previous tasks, ensuring that every aspect of the network and building design meets the desired standards and objectives. This phase involves the design and implementation of a comprehensive network infrastructure, including floor planning, selecting necessary devices, developing an IP addressing scheme, outlining connection layouts, and documenting the entire project. Through reflection and detailed documentation, this phase will provide an in-depth overview of the project's accomplishments and propose suggestions for future improvements.

6.2 Project background and an overview of the client's status and issue

Currently the Faculty of Computing (FC) has a total number of 1800 students (both undergraduate and postgraduate), 100 academic and 40 supporting staffs. Due to the anticipated 15% growth of both students and academic staffs in the next 4 years, FC is planning a renovation project by building a new 2-storey building to accommodate the increasing people. Below shows the list of required work areas in the building:

1. 2 General Purpose Lab
2. 1 Cisco Network Lab
3. 1 Embedded Lab
4. 1 Video Conferencing Room
5. 1 Hybrid Classroom
6. 1 Student Lounge

To begin with, each lab must be equipped with 30 workstations. They must be equipped with a fast connectivity networking infrastructure to prepare for education in line with 4IR. The Cisco Network Lab is used for education purpose equipped with high-end networking devices, including switches and routers. Next, the Embedded Lab should be equipped with devices and peripherals for best education experience. The Hybrid Classroom should also have a good setup to enhance teaching and learning experience. In order to provide best

comfortability, it is important to note that every lab sizes must be suitable for the student's size, the number of workstations and the total networking devices amount. Lastly, the student lounge allows students to work and relax in a comfortable environment while providing the best network services. FC Dean required that the building should be scalable, manageable, reliable and cost efficient. Hence the building should be equipped with better networking equipment that promotes growth, efficiency and security so that it can last for the next 20 years.

6.3 Compiled Solution for Task 1-5

Task 1: Project Setup

1.1 Brief Explanation for the Task

The Faculty of Computing (FC) is planning to construct a new two-storey building equipped with modern technology, such as networking devices that support education in line with 4IR standards, to accommodate an anticipated 15% growth in student and academic staff numbers over the next four years. Our group aims to design a cutting-edge and cost-effective network infrastructure that supports growth, ensuring students and staff have access to technology in a comfortable environment for study.

The building should include 2 general-purpose labs, 1 Cisco Network lab, 1 Embedded lab, and a student lounge, each 14m x 10m in size and a total 30 workstations for each lab with equipped a high-speed internet connection. We should design the floor plan of the building with any suitable software based on the requirements on the case study. The network design should feature a high-speed core backbone, secure segmentation, and protocols to protect against network breaches. Additionally, the infrastructure will be scalable and support wireless connectivity, supporting future growth and equipped with technological devices, in line with Faculty Computing's objective to create an innovative and comfortable learning environment.

1.2 Suggested Floor Plan

First Floor

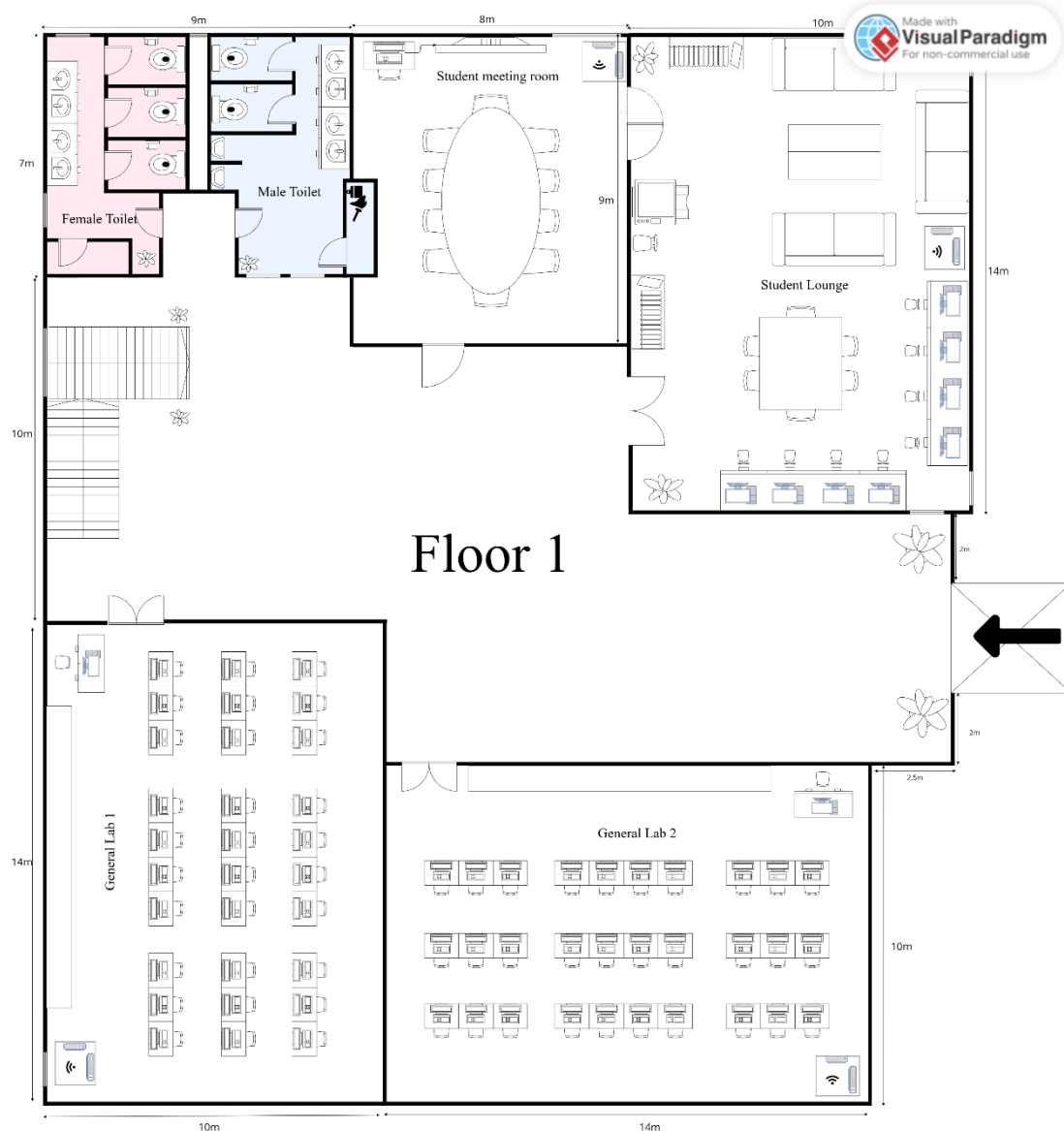


Figure 1.0

Figure 1.0 above shows the full layout of the first floor of the building. It is equipped with 2 general labs, 1 student lounge, 1 student meeting room and 2 toilets. The general labs are placed at first floor so that it's much easier for students to access. Each lab can accommodate a maximum number of 30 students and each table are equipped with a monitor and a CPU for enhanced learning experience. Both labs are also equipped with a high-bandwidth WIFI router to give a seamless and smooth network experience during lecture sessions. Both are width 10m and a length of 14m.

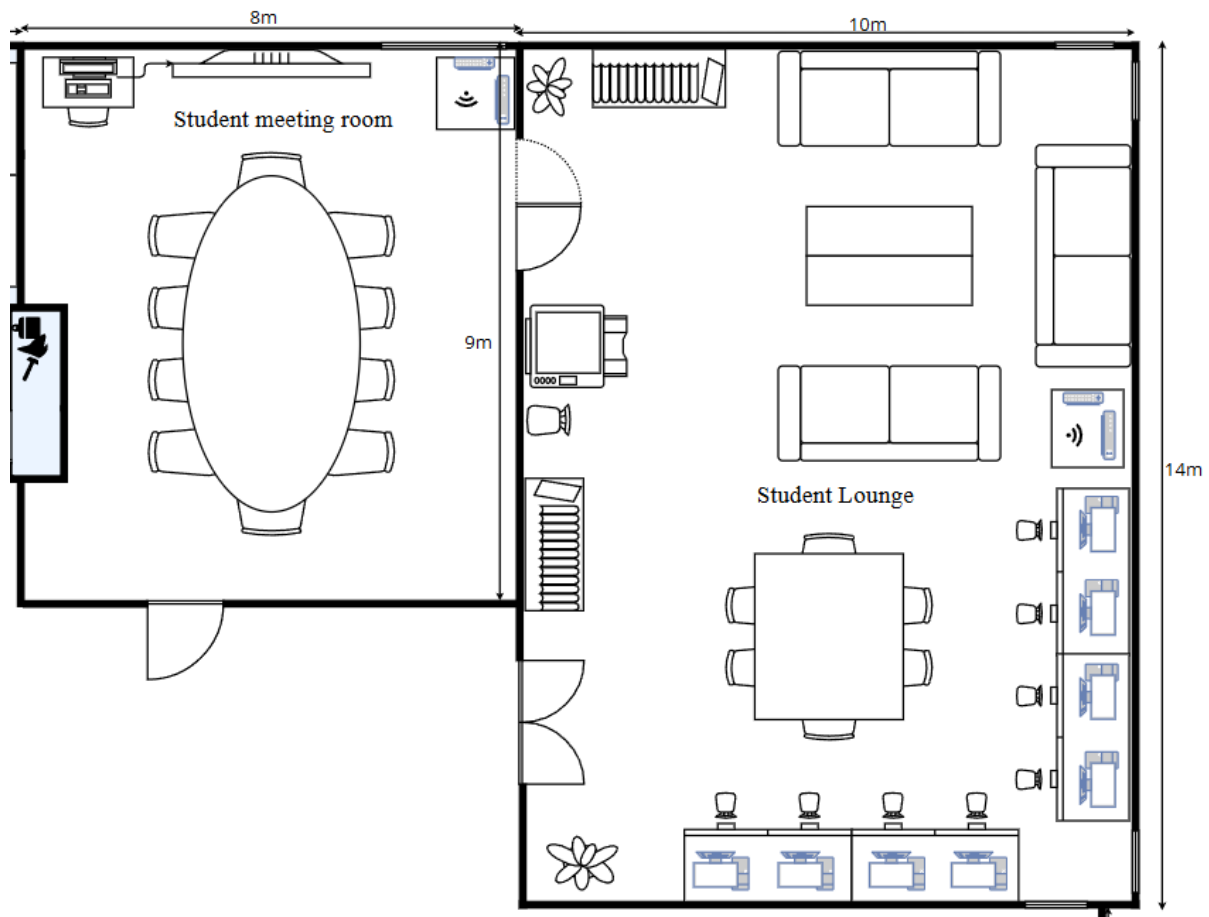


Figure 1.1

Figure 1.1 shows more details of the student lounge and meeting room. The student lounge provides students an area to work and relax while connected to the network via WIFI. The room is equipped with books and computers for students use. Beside it is a student meeting room which is accessible through the door from the hallway or student lounge. It aims to provide a suitable environment to discuss their group project or assignments. The student lounge measures 14 m by 10 m, while the meeting room is 9 m by 8m. We hope that these 2 facilities will provide a much more favourable environment for student to study or even rest.

Second Floor

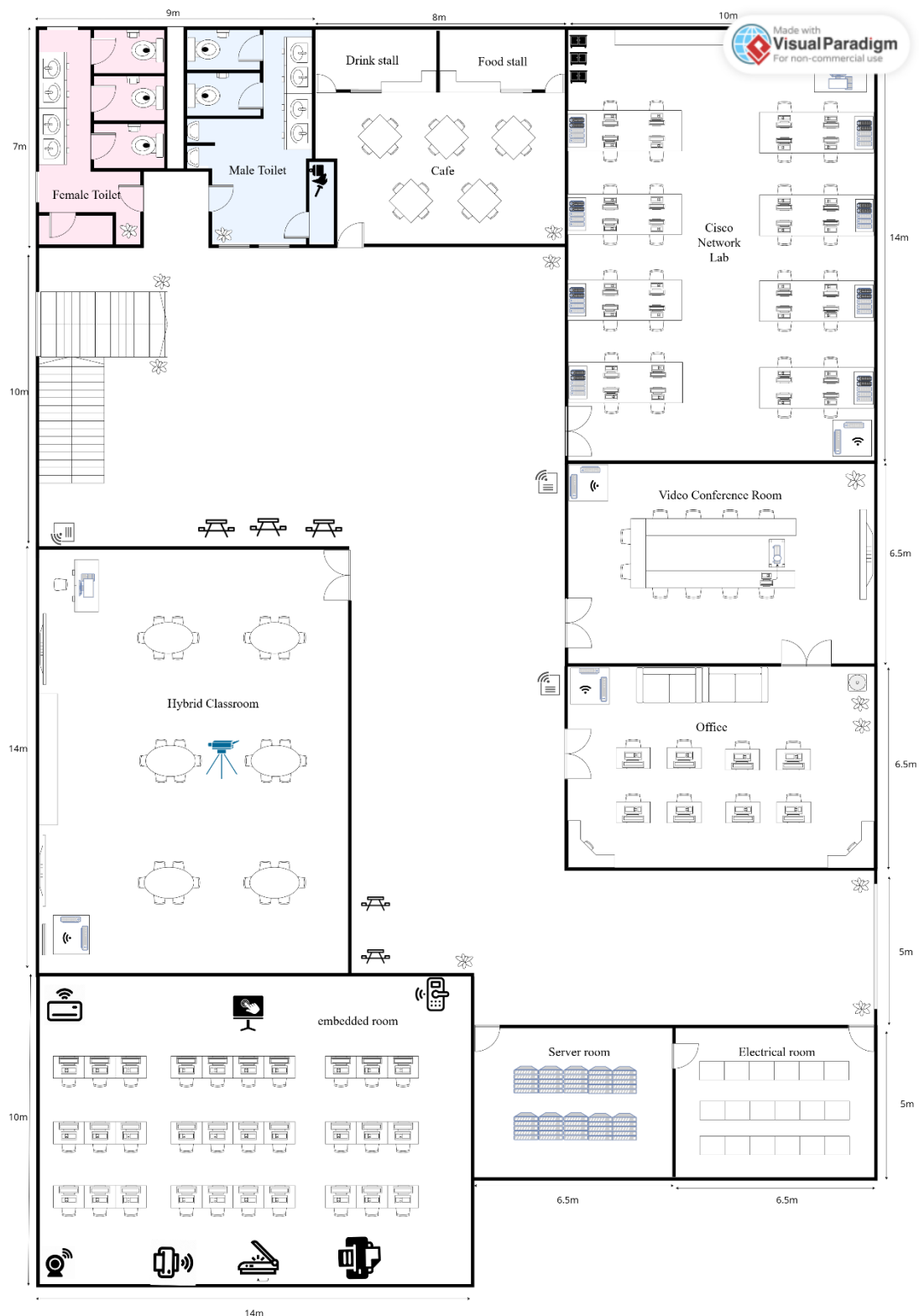


Figure 1.2

The second floor layout includes various rooms and facilities designed for accessibility and functionality. On the right side, there is a Cisco Network Lab (10m x 14m) with multiple workstations, monitors, and CPUs for student use, along with high-bandwidth

Wi-Fi for seamless connectivity. Adjacent to it is the Video Conference Room (6.5m x 10m) with a large table and monitors for virtual meetings, and next to that, an Office space (6.5m x 10m) with desks and computers for staff. The bottom left features the Embedded Room (14m x 10m) with specialized equipment for embedded systems, while the bottom right houses a Server Room (6.5m x 5m) for networking equipment and an Electrical Room (6.5m x 5m) for the building's electrical infrastructure. The left side includes a Hybrid Classroom (10m x 14m) with round tables for group work and hybrid learning equipment. Near the center, a cafe area with food and drink stalls offers seating, and there's an open student lounge with additional seating. Male and female toilets are located near the cafe area for convenience.

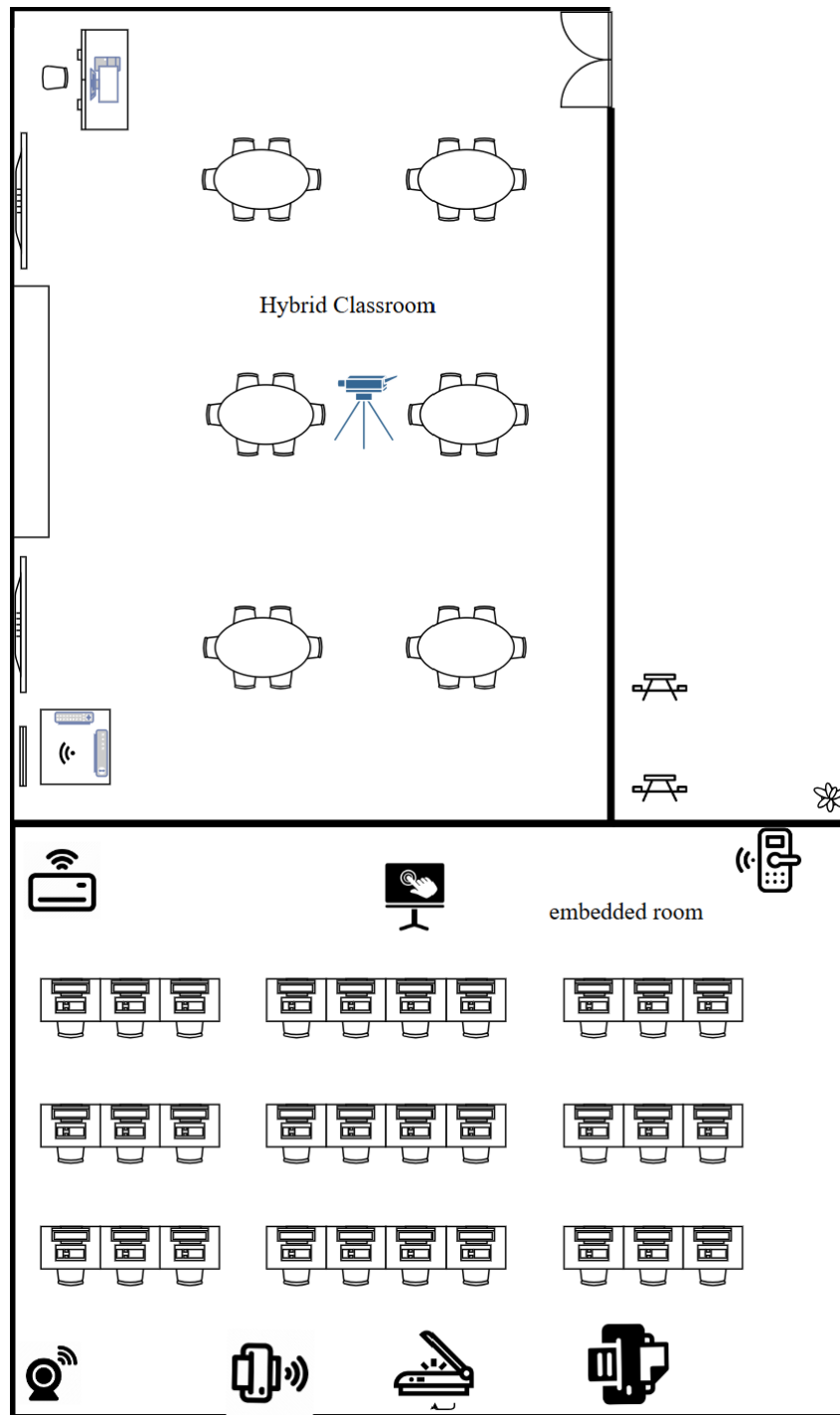


Figure 1.3

Figure 1.3 provides a detailed view of the Embedded Room and Hybrid Classroom. The Hybrid Classroom offers a conducive learning environment, featuring a 360-degree camera for monitoring students, two large screens at the front for video conferencing, and tables and chairs arranged in groups to facilitate discussions. High-speed Wi-Fi ensures smooth connectivity. The Embedded Room is equipped with IoT-enabled devices, including a wireless air conditioner, scanner, printer, and a digital camera. Access to the room is secured with a digital lock, with all technology in the room operated through IoT integration.

1.3 Brief Explanation for The Layout

This layout represents a modern academic facility tailored to the Faculty of Computing's needs, with a clear organization of spaces for both teaching and collaborative learning.

For the Central Features, the main entrance opens into a café area with drink and food stalls, providing a casual space for students and staff to relax or study informally. Nearby are the male and female restrooms, conveniently located for ease of access from different parts of the building. This café area serves as a social hub for students, fostering interaction and a sense of community.

The Academic and Collaboration space in on the right side, the layout includes a Cisco Network Lab, a dedicated space for networking activities and teaching. This lab is arranged to accommodate multiple workstations with adequate spacing, which is essential for hands-on networking exercises. Just below, there is a Video Conference Room equipped with a large table and conference technology, designed for virtual meetings and collaborations with remote participants.

The Hybrid Classroom on the left side of the layout is designed to support both in-person and remote learning. It features multiple round tables, likely intended for group discussions, collaborative projects, and interactive teaching methods. A hybrid setup, typically with a central display and conferencing equipment, is ideal for blended learning, enabling effective engagement for both on-site and remote students.

As for Administrative and Technical space which is towards the bottom of the layout, there is an Office space designated for staff or administrative work, equipped with desks and essential office furniture. Adjacent to the office, the Server Room and Electricity Room are located, centralizing the building's IT infrastructure and ensuring quick access for maintenance or troubleshooting. This setup ensures that network operations and electrical systems are secure and easily manageable, meeting the building's technical demands.

Overall, this layout efficiently combines academic, collaborative, and technical spaces to support a seamless educational experience, encouraging both learning and professional development while maintaining robust infrastructure for future growth.

1.4 Reflection of Task 1

In Task 1, we were supposed to design the layout for the first and second floors of an academic facility to meet the needs of the Faculty of Computing. The task included designing an area that will provide a comfortable atmosphere for academic learning and collaborative engagement while considering functional design for administrative and technical purposes.

Among the key things learned was how to strike a balance between academic, social, and technical spaces. Our designed layout has integrated various types of learning

environments, starting from the Cisco Network Lab and Hybrid Classroom to the Video Conference Room, which allow for both face-to-face and distance learning in favor of various teaching methodologies and provide a greater degree of flexibility for students and instructors alike. We also made sure areas like the café, strategic locations for restrooms, and collaboration spaces facilitate social interaction and a sense of community—all ingredients of a positive learning environment.

We did have some space allocation challenges in trying to balance the academic needs with the technical ones in the facility. The location of the Server Room and Electricity Room was crucially important in maintaining the building's IT infrastructure and network operations, teaching me the importance of accessibility and security regarding managing a building's technical needs. Similarly, the design of the office space and other administrative areas had to be made with care to ensure smooth operations without interfering with the learning spaces.

In summary, this project deepened my understanding of space planning, user experience, and how educational facilities can be designed to address current needs while anticipating future growth. I considered the relationships between space types in order to create a smooth, functional, inclusive academic facility.

Task 2: Initial Design – Preliminary Analysis

2.1 Question

- 1. What are the specific expectations for wireless connectivity and range coverage across both floors?**

The wireless connectivity must provide uninterrupted access to the building, including all labs, student lounges, and common areas. Cisco Network Lab and Hybrid Classroom will host bandwidth-intensive activities such as online conferencing and real-time collaboration, it required a high bandwidth and speed of Wi-Fi with minimal. Therefore, the Wi-Fi routers should place at least one per room because to eliminate dead zones problem and ensure reliable connectivity for all users. These routers should be chosen to accommodate high-density usage, especially in labs where multiple devices connect simultaneously.

- 2. What devices and functionalities are required for the Cisco Network Lab?**

For an ideal setup in the Cisco Network Lab, the recommended equipment is two cisco catalyst 3750 switches [1]. These switches provide Layer 3 routing capabilities and are excellent for teaching network switching and routing protocols. They support VLAN configurations, enabling students to learn network segmentation and traffic management in a controlled environment. Another recommended equipment is cisco 2800 series routers [1]. These routers support basic routing protocols and WAN connections which are vital for understanding and configuring network communication across different networks. The Cisco 2800 routers also support multiple interfaces which allow students to connect different network segments.

- 3. What specific security protocols should be implemented to protect against common threats like DDoS attacks and internet worms?**

First, firewalls should be deployed at key network entry points to monitor and control incoming and outgoing traffic based on predetermined security rules [2].

Intrusion Detection and Prevention Systems (IDPS) are also essential, as they both passively monitor traffic and actively block suspicious or malicious behaviour once flagged. Second, network segmentation divides the network into secure segments or VLANs based on different user roles, such as student and staff access [3]. This limits access within the network and minimizes the impact of a breach in one segment from affecting the entire network. Third, we can implement the 802.1X Authentication combined with Radius Server [4]. This protocol enforces network access control by requiring users to authenticate with UTM student ID and password before connecting to the Wi-Fi. It can secure the student privacy since 802.1X authentication protects people to walk in and release virus, malware or man-in-the-middle attack.

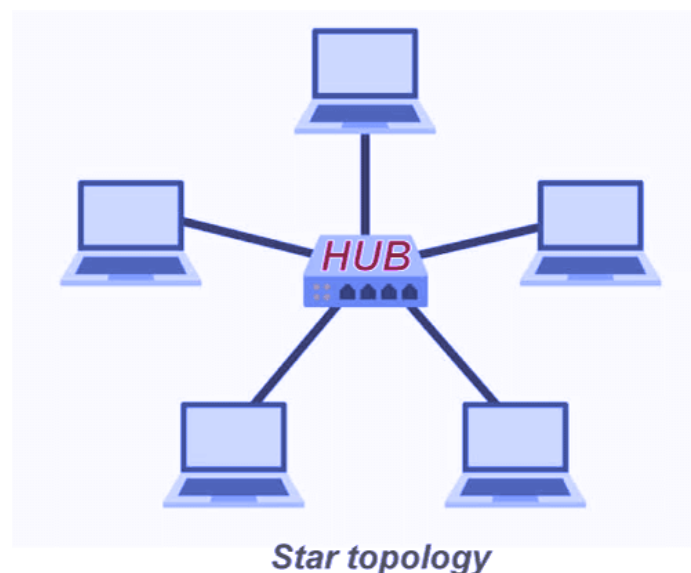


Figure 2.1 Star topology

4. What network topology would be suitable inside the labs?

The network topology used varies in many aspects, especially since the faculty building contains two floors and several different labs. In our group opinion, the most suitable network topology for the labs in this project is the Star Topology. In a star topology, all devices such as workstation and IoT devices are connected to a central hub or switch [5]. This central hub allows for easier network management, as problem can be isolated and resolved quickly without affecting the entire lab network. If one connection fails, it does not affect the others, making it more reliable.

Furthermore, each device has a direct link to the switch, there is minimal network congestion which resulting lower latency and faster data transmission speeds. This is crucial for lab environments, especially the Cisco Network Lab and Embedded Lab, where high-speed internet and seamless connectivity are required for educational activities.

5. What is the minimum bandwidth for each room?

There are a total of 4 labs, namely 2 general-purpose lab, 1 Cisco Network lab and 1 Embedded lab. We plan to install an internet router with a minimum 30Mbps network speed per device. This is to ensure overall smooth experience in preparation for education in line with 4IR, which includes IoT, AI-driven applications, big data, and real-time analytics. We also consider the large number of students (30 students per lab) when assessing suitable internet speeds, as high network usage from activities like streaming and resource-intensive applications (e.g., Cisco Webex meetings) can place significant demands on bandwidth. This totals up to a minimum 1Gbps per lab.

Besides, the video conference room needs a higher internet bandwidth. Since it is only equipped with a Smart TV for online meeting purposes, it will be connected through ethernet to the router for a more stable connection. Additionally, this setup minimizes interference and enhances security by creating a more direct, controlled connection to the network. Besides, taking into consideration that the streaming quality and audio quality should be top notch, we plan to install a minimum 100Mbps dedicated router for the room.

Moreover, there is a hybrid classroom which require simultaneous high-quality video streaming and interactive platforms (e.g., real-time collaboration tools). The video streaming platform should provide high image and audio quality. Next taking the total number of students into account, the network would need to handle multiple connected devices. In conclusion, with 30 Mbps provided for each device, the total bandwidth for hybrid classroom should be 1Gbps.

The new building also includes new facilities such as student lounge and a student meeting room. We assume that both rooms are equipped for students to do normal daily tasks such as web surfing and online meeting, hence a minimum 10

Mbps should suffice for each device. Since there could be lots of students staying in both rooms, the total bandwidth should be increased to provide adequate speeds that can support multiple users. There's also a working office which can accommodate up to 8 lecturers. If lecturer also handle web browsing and online meetings, a minimum 3 Mbps should suffice for each device too. To conclude, a total of 100Mbps for student lounge and student meeting room each, and total of 24Mbps for the office.

Lastly, the new building is also equipped with a server room. It's a small-scale server room used to support basic file storage, internal applications, and email servers. With that in mind the bandwidth speed should be at least 100Mbps. This ensures smooth performance for daily tasks and prevents congestion, while also allowing for future scalability as the network demands grow.

6. What type of networking cable is suitable to connect to the computers? Why did you choose it?

Category 6 Ethernet cable would be a suitable choice for connecting the computers. Category 6 cables are a type of network cable used for connecting devices to a local area network (LAN) because they offer a good balance between speed, distance, and cost [6]. With a bandwidth capacity of up to 250 MHz, Category 6 cables can support data transfer rates of up to 1 Gbps (Gigabit Ethernet) over distances up to 100 meters, which is ideal for most classroom, office, and lab environments shown in our design [7]. This makes it appropriate for handling the data requirements of labs, classrooms, and conference rooms, where multiple devices may need to transfer large amounts of data simultaneously. We choose this cable as it fits all the needs for each machine to communicate and connect to the Internet seamlessly.

The structured layout of our floor plans suggests spaces where a stable and high-speed network connection is essential, such as in the Cisco Network Lab, General Labs, and the Hybrid Classroom. Category 6 cables are designed to reduce crosstalk and interference, which is especially important in high-density environments where numerous cables may run close to each other [7]. This cable type will ensure a reliable connection for video conferencing, streaming, and network-dependent

software used in collaborative spaces, like the video conference room and the student lounge area.

In addition, using Category 6 cables allows for future scalability. If network demands increase, Cat6 can support 10 Gigabit Ethernet over shorter distances (up to 55 meters), which could be beneficial if Faculty Computing plans to upgrade to higher-speed networks. For areas with high traffic, such as the server room or labs, this extra capacity could be valuable. Overall, Category 6 Ethernet cables offer the best combination of performance, cost-effectiveness, and futureproofing, making them a suitable choice for the networking needs reflected in our design.

7. How to ensure that the coverage of wireless networks can support the area of the faculty?

In our opinion, there are several ways to ensure that the coverage of wireless networks can support the area of the faculty. First, choosing a central location at each floor for the router can improve the coverage of wireless networks as a router in the central position provides the perfect signal coverage for the entire building.

Besides, adding a wireless repeater is also a good way to ensure the coverage of wireless networks. Wireless repeater can rebroadcast a wireless signal and strengthen the signal from the router to other floors of the building. We can place the wireless repeaters at the classrooms or labs that receives weak wireless signals so that the performance of the network in those rooms can be improved.

Apart from that, we can also use an ethernet cable to connect the access point to the router by inserting the cable into a LAN port on the router. This introduces the access point to the router's created local area network then connects the access point to the computer with an ethernet cable so that the computer can direct access to access point and LAN management. Once the wireless access point is done, students can enjoy expanded wireless coverage and improved wireless signal. The issues that could arise when dealing with coverage is that the maximum number of clients that can be served by a single access point highly depends on the fact that what internet services are being availed by the users. Then, sometimes users can associate with an access

point located in a distance area instead of connecting to a nearby access point which is offering weak signal strength due to co-channel interference or interference caused by walls or other solid substances [5].

8. Is there a preference for wired or wireless connectivity in different areas of the building, such as labs versus the student lounge?

Yes, there is a distinct preference for different types of connectivity based on the specific requirements of each area in the building. In the general-purpose labs, Cisco Network Lab, and Embedded Lab, wired connectivity is preferred. This is because wired connections provide more stable, high-speed, and low-latency internet access, which is essential for activities like running network simulations, IoT experiments and high-bandwidth applications. Wired connections also offer better security and reliability compared to wireless, making them ideal for labs where consistent performance is crucial.

In areas like the student lounge, meeting rooms, and cafe, wireless connectivity is preferred. These are spaces where flexibility and mobility are prioritized, allowing students and staff to connect their laptops, smartphones, and tablets without being constrained by physical cables. Wi-Fi access points provide convenient, high-speed internet access for users to study, collaborate on projects, or relax while staying connected.

9. What are the key elements in a hybrid classroom?

In a hybrid classroom, a reliable high-speed internet connection is the foundational network element that ensures seamless connectivity for both in-person and remote participants. This robust connection supports online resources, live video conferencing, and interactive platforms essential for hybrid learning. Since multiple users, both in the classroom and online, will be accessing resources simultaneously, a strong internet connection prevents lag or disruptions and enables real-time collaboration. Typically, a wired or fibre-optic internet link is preferred for its stability and faster speeds compared to wireless options.

Wireless Access Points (WAPs) are crucial in a hybrid classroom to provide Wi-Fi coverage throughout the room. While wired connections are preferred for devices that require constant, high-speed connectivity, WAPs enable students and instructors to connect laptops, tablets, or other devices wirelessly. A strong Wi-Fi signal ensures flexibility in seating arrangements and smooth access to online resources, facilitating effective participation for in-person students while interacting with remote learners.

Ethernet cabling, such as Category 6 (Cat6) Ethernet cables, is essential for providing stable, high-speed connections to key devices like the instructor's computer, streaming equipment, and interactive displays. These wired connections reduce latency and improve performance for high-bandwidth activities like live streaming and video conferencing. A wired setup minimizes the chances of connectivity issues that may affect the quality of audio and video feeds during hybrid sessions, making it an essential component of a reliable hybrid classroom.

A network switch connects multiple devices within the classroom to the local area network (LAN) and efficiently manages network traffic between them. The switch links devices such as computers, streaming equipment, displays, and possibly Internet of Things (IoT) devices, enabling seamless communication across the network. By serving as a central point for network management, the switch prevents bottlenecks and ensures that each device receives the necessary bandwidth, even during periods of high network usage.

Finally, firewall and security solutions are critical to safeguard the hybrid classroom network. With multiple devices and participants connected, strong cybersecurity measures prevent unauthorized access and protect sensitive data. A firewall monitors and filters incoming and outgoing network traffic, ensuring data privacy and network safety. This secure environment allows instructors and students to focus on learning without concerns about potential cybersecurity threats.

10. How do you plan to reduce Internet service downtime issues?

We acknowledge that network downtime poses a significant threat as it disrupts the flow of the class and online meetings, leaving multiple users with an unsatisfactory learning environment. First, we plan to install a backup router and network switches to deal with primary device failure. These can be automatically switched over without affecting the network, thus ensuring no interruption to online activities.

Besides, we can prevent downtime issue by installing multiple ISPs to the new building [8]. These ensure that whenever one ISP encounters slow speed or outage, the other ISP will immediately take over, hence reducing network delay in the process. Both measures utilise the server room that aim to solve the problem from the aspect of network redundancy, making it more cost effective.

For extra protective measure, we plan to install proactive network monitoring tools such as SolarWinds, Auvik and Datadog. These tools are tasked to detect any network traffic irregularities and connectivity issues, hence giving IT staff more time to solve the issues.[9]

11. What budget constraints should be considered when choosing networking devices and equipment?

When selecting networking devices and equipment for the project, several key budget constraints need to be considered, starting with the initial equipment costs. The primary expenses include the cost of networking devices such as workstations, switches, routers, access points, firewalls, and IoT devices for the Embedded Lab and Cisco Network Lab.

In addition, the building needs to accommodate 1,940 people, including students and staff, with an expected 15% growth over the next 4 years. Given this projected increase, it is crucial to select high-performance devices that can support future needs with cutting-edge technology, without requiring a complete overhaul. Although scalable devices may have higher upfront costs, they can save money in the long run by reducing the need for frequent upgrades.

Lastly, maintenance and support costs must also be considered within the budget. The total cost of network devices includes not only the initial purchase but

also ongoing maintenance, software updates, and technical support. It is more preferable to choose devices with extended warranty coverage and reliable customer service, which can help minimize unexpected expenses. Additionally, premium brands often come with higher support costs, which should be factored into the overall budget.

2.2 Feasibility Assessment

After answering all the questions, we have gained a clear understanding of the project requirements. We have evaluated the feasibility assessment from three key aspects: economic feasibility, legal feasibility, and technical feasibility.

Economic feasibility

The project's original budget includes expenses for networking equipment such as workstations, switches, routers, access points, and cables. Since the number of students and employees is predicted to increase by 15% over the next five years, equipment selection needs to find a balance between cost-effectiveness and performance requirements in order to maintain long-term financial viability. Although they may cost higher up front, high-performance devices like Cisco switches provide more scalability and dependability, which reduces the need for regular upgrades. Besides that, we can select devices with extended warranty coverage which will help reduce unexpected expenses, maintenance, technical support and software upgrades. Additionally, the project should include a contingency budget for unexpected expenses, such as additional equipment needs or network upgrades, ensuring the project remains cost effective even if updates are necessary.

Technical feasibility

Technical feasibility studies ensure we have the technical resources needed to meet project requirements and turn ideas into results. This study confirms that required hardware and software components can be sourced affordably from reliable vendors like Huawei and Cisco, with official purchase options. Subscription-based software, such as Microsoft and antivirus programs, can be negotiated with providers. These hardware and software will ensure our technology is able to keep up with the current building development, making the process easier to handle. Additionally, a minimum of 5 IT staff members are required to manage equipment and server maintenance, ensuring the technical feasibility of the new building."

Legal feasibility

This is significant as a procedure to avoid any potential issues from a legal perspective. It is best to avoid legal rights and copyrights of the software used for the network equipment such as Microsoft Office and so on. Therefore, all the licensed software will be purchased through the legal pathway that we need to hand on to the Department of Digital Services (UTMDigital) to deal with. Furthermore, we also need to ensure the purchase of hardware is following the original and legal channel. To avoid legal issues, all hardware and software will be sourced from certified suppliers. UTMDigital will oversee licensed software, including UTM Webex and Microsoft products, ensuring compliance with regulations and preventing the use of pirated goods.

2.3 Reflection of Task 2

In this task 2, we recognised the importance of conducting a Preliminary Analysis before initiating any project. We understand the feasibility of the proposed network infrastructure for the Faculty of Computing after this task. Through this analysis, I learned how to evaluate economic, technical, and legal aspects to ensure the project's success.

Task 3: LAN Device Analysis

3.1 Introduction list of devices uses

1. Routers

A router is a device designed to link two or more packet-switched networks or subnetworks. Its main roles include directing data packets to their correct IP addresses and enabling multiple devices to share a single Internet connection [1]. While there are various types of routers, most are used to transmit data between local area networks (LANs) and wide area networks (WANs). A LAN refers to a network of interconnected devices within a limited geographical area and typically requires only one router [1]. The key functions of a router include network traffic management and IP address assignment. Firstly, a router manages network traffic by routing data between various devices, such as computers, smartphones, and printers, within a local network as well as between the local network and the internet. Secondly, it handles IP address assignment by using the Dynamic Host Configuration Protocol (DHCP) to assign unique IP addresses to each device on the network, ensuring smooth and efficient communication.

2. Switches

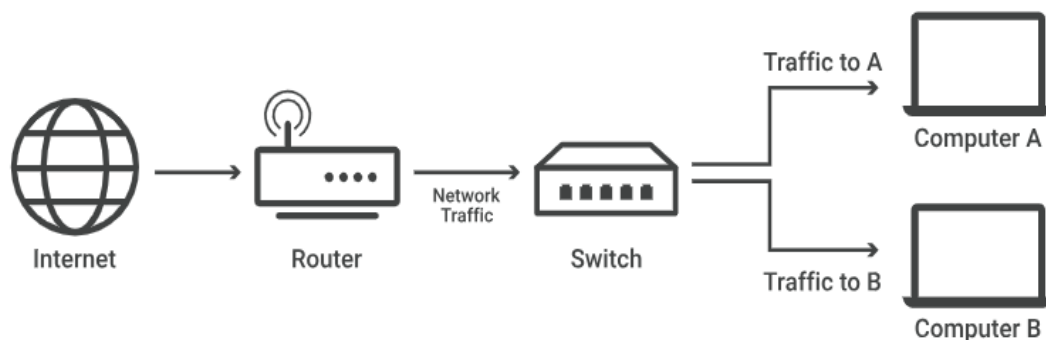


Figure 3.1.1

A network switch links devices within a network, allowing them to communicate by exchanging data packets [2]. Switches connect various devices, including computers, wireless access points, printers, and servers, within a building or campus. They are crucial parts of any network. The switch facilitates information sharing and effective communication between these linked devices [3].

3. Patch panels

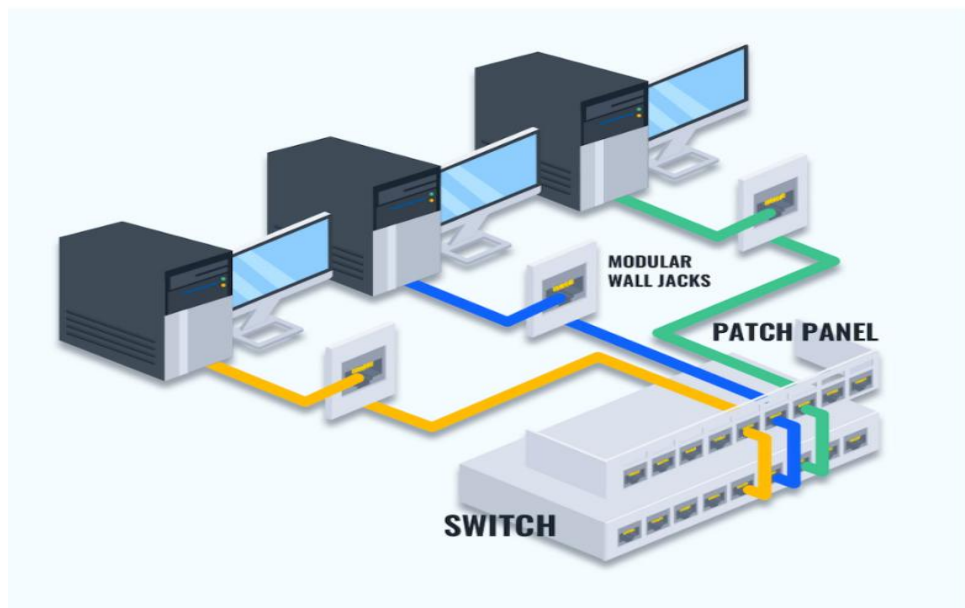


Figure 3.1.2

A patch panel is a networking device that has several ports for connecting different Ethernet cables [4]. It simplifies cable management by offering a single spot for classifying and marking every port [4]. Additionally, patch panels facilitate the separation of several networks and enable quick troubleshooting of any difficulties that may develop. For instance, fiber -optic patch panels are widely used patch panel types that are perfect for supporting a variety of optical fiber connections since they are specialized in using light signals to transmit data and images over thin optical fiber [4].

4. Ethernet cables

An Ethernet cable is a type of networking cable that connects computers, routers, switches, and other network devices to a local area network (LAN) or the Internet. It is mostly utilized for wired data transmission, providing a steady and dependable connection for fast data transfers. Multiple twisted pairs of copper wires are commonly seen in Ethernet cables, which serve to increase signal quality and lower interference. varying types of Ethernet cables (such as Cat5e, Cat6, Cat6a, Cat7, and Cat8) have varying frequencies and maximum data transfer speeds. Cables in higher categories (such as Cat6 and above) can travel farther and quicker.

5. Wireless access point

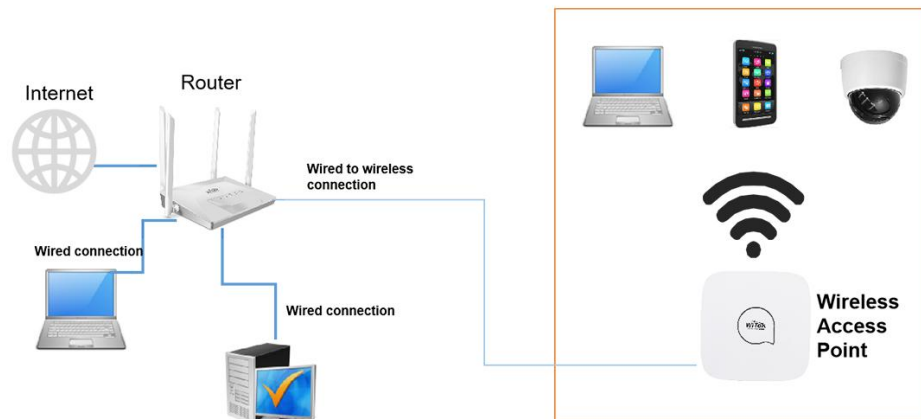


Figure 3.1.3

Wireless access point is a networking device that connects devices with wireless-networking capabilities to access the wired network given that they must be within connection range of the wireless AP. This includes routers, switches, endpoint devices etc. Wireless AP can connect the clients to the network using specific frequencies. However, the range covered can be influenced by factors such as its placement, surrounding obstacles, active devices frequency disruption and others, hence wireless AP can also function as a signal range extender to provide wider wireless coverage [5]. What makes wireless AP stands out from wired AP is that its more convenient to connect all the network devices to the network wirelessly rather than using physical wires and cables.

6. Fiber optics

Fiber optics is a recent technology that act as a medium to allow data transmission in the form of light pulses over long distances at a high speed. It's primarily made from silica glass for better refraction of light. What makes fiber optics stand out from coaxial Internet cables is its reliability during power outage or electricity interruption. As fiber optics uses light to transmit data instead of electricity, its operational function won't be disrupted hence making it more favourable in these situations [6]. Fiber optic Internet promotes higher bandwidth and can reach upload and download speeds of a range of at least 250 to 1,000 Mbps [6]. It has significantly improved the Internet connection speed, making it more reliable in high density urban areas or working environments where fast and reliable connection is substantial.

3.2 Table of Devices

Devices	Description	Price per unit (RM)	Quantity	Total price (RM)
Router  Figure 3.2.1	TP-Link Archer AX90 (AX6600) tri-band Wi-Fi 6 setup for fast, reliable wireless connectivity. It supports speeds up to 4804 Mbps on 5 GHz for high-speed applications, 1201 Mbps on a secondary 5 GHz band, and 574 Mbps on 2.4 GHz, providing flexibility for various usage scenarios Ultra Connectivity: Equipped with one 2.5 Gbps WAN/LAN port and one 1 Gbps WAN/LAN port, along with three additional Gigabit LAN ports and two USB ports for media sharing and storage, it maximizes connectivity options Coverage and Performance: Features eight high-gain antennas with Beamforming technology to ensure extended coverage, while OFDMA and MU-MIMO allow for low-latency connections to multiple devices simultaneously Processor: Powered by a 1.5 GHz quad-core processor, the Archer AX90 ensures fast data processing for smooth operation in busy networks	799.00	29	23171
Switches  Figure 3.2.2	TP-Link TL-SG3428X High-Speed Connectivity: The switch has 24 Gigabit RJ45 ports that ensure reliable, high-speed connections for connected devices, which is essential for smooth data flow in a building environment	1445.00	24	34680

	10G Uplink Ports: It includes 4 × 10 Gbps SFP+ slots for uplinks, allowing high-bandwidth connections to the core network or other switches. This is critical for avoiding bottlenecks, especially in large networks with heavy traffic Advanced Security: Offers IP-MAC-Port Binding, Access Control Lists (ACL), DoS protection, 802.1X, Radius Authentication, DHCP Snooping, and more. These security measures help protect the network from unauthorized access and attacks, which is important in environments with sensitive information Management Options: Integrated into Omada SDN for centralized management, as well as standalone management through Web UI, CLI, SNMP, RMON, and dual image support, making it versatile and easy to control Quality of Service (QoS): Supports L2/L3/L4 QoS and IGMP snooping to optimize voice and video applications, ensuring efficient use of bandwidth for critical services			
Patch panels  Figure 3.2.3	D-Link NPP-C61BLK24 (24-port CAT6) Category: Cat6, supporting up to 10 Gbps network speeds, suitable for most modern networking needs and helps organize Ethernet cabling in a structured and easily maintainable format Port Count: 24 RJ45 ports, it is suitable for multiple connections across labs, faculty offices, and shared spaces, providing high-speed data transfer rates essential for academic and administrative applications.	178.88	6	1073.28

	Mounting: Standard 19-inch rack-mountable design, ideal for server racks and network cabinets Durability: Constructed from high-quality metal with a solid build that ensures longevity in high-usage environments Cable Management: Integrated rear cable management features help maintain neat, organized cabling and prevent tangling			
Ethernet cables  Figure 3.2.4	UGREEN Cat6 Ethernet Cable Data Speed: Transmits at speeds up to 1000 Mbps (1 Gbps), making it ideal for high-speed applications like cloud computing and HD video streaming. Bandwidth: Provides a 250 MHz frequency, ensuring better stability compared to Cat5 and Cat5e cables. Durability: Equipped with a flexible PVC outer jacket for protection and long-lasting use. Connectors: Features gold-plated RJ45 connectors, ensuring corrosion resistance and optimal data transfer performance. Ease of Installation: Its design is ideal for easy installation in office or institutional environments.	9.85/2 m	250 m	2462.5
Wireless access point	Tp-Link AX3600-EAP660 HD Ultra-Fast Wi-Fi 6 Speeds: 1148Mbps on 2.4GHz and 2402Mbps on 5GHz Wireless Standards: IEEE 802.11ax, also known as Wi-Fi 6 ensuring capacity, performance and efficiency	1758.00	10	17580

 Figure 3.2.5	Frequency Band Class: Dual band, where connection can be done through two different frequencies which aid in overall network performance High-Density Connectivity: 4x increased capacity, meaning higher throughput median in dense environment compared to 4x4 11ac access point Centralized Management: Omada Software Defined Networking platform, singularizing all access points configuration into one, also enhancing security, efficiency and reliability Seamless Roaming: Unaffected video streams as users move between floors, maintaining a consistent connection without buffering or disruptions Placement: Wall / Ceiling mount Feature: Zero-Touch Provisioning (ZTP)[§] to reduce manual intervention, allowing automatic configuration and deployment of networking devices efficiently			
Fiber optics  Figure 3.2.6	OM4-LC/UPC-LC/UPC-DX-1M Fiber Optic Fiber Mode: OM4 50/125µm, providing a higher bandwidth up to 4700 MHz/km bandwidth at a wavelength of 850 nm. Also allow longer reach of up to 550meters at 10Gbps, and up to 150 meters for 40 Gbps and 100 Gbps transmissions. This ensures high data transmission between short to medium range.	20.00/1 m	486.5m	9730

	Fiber Count: 2 Fibers, ideal for simplex or duplex links, providing reliable high data transmission Type: LC UPC Duplex, allowing high-density applications while minimizing signal reflection for a better network connection performance Cable Type: Tight Buffered, protecting every individual fibre from physical stress Jacket Rating: OFNR (Riser), flame retardant materials to reduce fire widespread along the cable Wavelength: 850/1300nm, versatile for different distance and ensuring best data transmission performance Temperature Range: -20 to 70°C			
Total:				RM886 95.78

3.3 Reflections

3.3.1 Are you surprised by the prices? How were you surprised?

Yes, the cost of networking equipment corresponds to its advanced features and technological capabilities. After discussions with the team, we finalized the quantities needed for each networking device, which resulted in a total cost of **RM88,695.78**. This amount constitutes approximately 10% of our overall budget. However, this estimate excludes the costs associated with the power station, building maintenance, furniture, decorations, and unforeseen expenses. Some devices, such as the TP-Link TL-SG3428X switch at RM1,445 and the Tp-Link AX3600-EAP660 HD wireless access point at RM1,758, appear to be reasonably priced, whilst others, such as the TP-Link Archer AX90 router at RM799.

What surprises us is how intricate and advanced technology these devices include. The switch includes advanced security features such as IP-MAC-Port Binding and DoS prevention, the router includes tri-band Wi-Fi 6 with a 1.5 GHz quad-core processor, and the wireless access point provides lightning-fast Wi-Fi 6 speeds with centralized management. We believe some hidden variables may also influence the price such as a brand's reputation, driving them to achieve higher prices in the market. While the price may seem high at first, they represent the complex networking devices needed to operate efficiently and effectively in an IT building, making the prices seems more justifiable. Some networking devices also provide with better security measures, for example FS OM4-LC/UPC-LC/UPC-DX-1M Fiber Optic's ability to prevent fire widespread through the cables. These features suggest cost-efficient methods to ensure the building durability for the next 20 years.

3.3.2 Have you ever considered cost as a factor for choosing networking devices?

Yes, when selecting networking equipment, price is an important consideration. We prior technical standards, compatibility, and dependability of the devices but it is crucial to balance them with financial limitations. For instance, certain high-end devices could include advanced features that are unnecessary for the requirements of a particular network, thus a less expensive alternative is more suitable and cost-effective.

In our project, cost considerations include the initial cost of the device, ongoing costs, scalability, and the trade-off between cost and performance. We must be able to afford the initial purchase price, but we also need to consider long-term expenses like longevity, maintenance, and energy efficiency since equipment with greater initial prices may end up saving money over time. A crucial component is scalability, so more costly devices with greater capacity may accommodate future network expansions without requiring urgent updates. Finally, brand and warranty are important since, even if they are more expensive, premium brands often provide superior warranties and assistance, reducing the need for unforeseen repairs or replacements.

3.3.3 What are the major differences between the same devices from different brands? For example, Cisco and Huawei Routers.

1) Router

For the router used in this project, we have chosen the TP-Link Archer AX90 (AX6600) router. The major difference between TP-Link Archer AX90 (AX6600) with other different brands we can compare is **Asus RT-AX86U**.

Specification	Asus RT-AX86U	TP-Link Archer AX90 (AX6600)
Max Wi-Fi Speed	Up to 5700 Mbps (Dual-Band)	Up to 6600 Mbps (Tri-Band)
Feature	Dual-Band, Advanced QoS, AiMesh	Tri-Band, HomeShield, 2.5Gbps Port
Security	Lifetime free ASUS AiProtection Pro, powered by Trend Micro™	HomeShield Basic
Price (RM)	1409	799.00

The Asus RT-AX86U and TP-Link Archer AX90 (AX6600) have varied features that are appropriate for varying budgets and purposes. The TP-Link Archer AX90 offers up to 6600 Mbps with a tri-band configuration, adding another band for better performance with numerous devices, while the Asus RT-AX86U has a maximum WiFi speed of up to 5700 Mbps with a dual-band arrangement. The Asus is perfect for maximum customisation and coverage extension because of its features, which include Advanced QoS and AiMesh for mesh networking capabilities. In contrast, the TP-Link offers a 2.5Gbps WAN port and HomeShield to meet the needs of consumers who want fast internet connections with minimal security measures.

The Asus router comes with a strong, integrated security feature called AiProtection Pro, which is powered by Trend Micro and is free for life. In comparison to Asus's product, TP-Link's HomeShield Basic provides basic security but less sophisticated safeguards. Finally, the TP-Link Archer AX90 is a more cost-effective choice, coming in at around RM799.

2) Switches

For the switches used in this project, we have chosen the TP-Link TL-SG3428X switches. The major difference between TP-Link TL-SG3428X with other different brands we can compare is **D-Link DGS-1210-28MP**.

Specification	D-Link DGS-1210-28MP	TP-Link TL-SG3428X
Ports	24 x Gigabit PoE, 4 x GbE/SFP combo	24 x Gigabit RJ45, 4 x 10G SFP+
Uplink Capability	4 x GbE/SFP combo ports	4 x 10G SFP+ for high-speed uplinks
Switching Features	L2+ switching, static routing	L2+ switching, L3 QoS, IGMP snooping
Security Features	Advanced L2 security features	IP-MAC-Port Binding, ACL, DoS, 802.1X, RADIUS
Price	2284	1445

Both the D-Link DGS-1210-28MP and the TP-Link TL-SG3428X are Layer 2+ switches that have unique characteristics to meet various networking requirements. The D-Link model is perfect for networks that need to provide power to devices like IP cameras and access points since it has 24 Gigabit PoE ports, 4 GbE/SFP combination connections, and a substantial 370W PoE budget. Additionally, it incorporates L2+ switching with static routing for fundamental network security and segmentation.

The TP-Link TL-SG3428X, on the other hand, has four fast 10G SFP+ uplink ports and 24 gigabit RJ45 ports, which increase bandwidth capacity for networks with massive data traffic. It also has advanced safeguards that improve network safety for critical situations, such as Radius Authentication, DoS prevention, and IP-MAC-Port Binding. The TP-Link model, which costs around RM1445, is more affordable than the D-Link, which costs RM2284.

3) Patch panel

For the patch panel used in this project, we have chosen the D-Link NPP-C61BLK24 (24-port CAT6) patch panel. The major difference between D-Link NPP-C61BLK24 (24-port CAT6) with other different brands we can compare is **Panduit CPP24FMWBLY 24-Port CAT6 Patch Panel**.

Specification	Panduit CPP24FMWBLY 24-Port CAT6 Patch Panel	D-Link NPP-C61BLK24 (24-port CAT6)
Port Count	24 x CAT6 ports	24 x CAT6 ports
Construction	Heavy-duty construction for industrial use	Durable metal body
Labeling and Identification	Enhanced cable management options	Basic cable management
Labeling and Identification	Clear labelling with color-coded port icons	Numbered ports for easy organization
Target Environment	Enterprise-grade, suitable for high-density networks	Small to medium-sized setups
Price	368.36	178.88

Panduit Patch Panel and the D-Link Patch Panel both include 24 CAT6 ports to meet various networking requirements. The Panduit panel is designed for enterprise-level applications, with heavy-duty construction, improved cable management options, and colour-coded markings for simple port identification, making it perfect for high-density configurations. The D-Link panel, on the other hand, is ideal for small to medium-sized networks, with a sturdy metal shell, basic cable organisation, and straightforwardly numbered ports for simple organisation. Within our project's given budget, the D-Link NPP-C61BLK24 is a suitable choice as it provides a practical and economical solution.

4) Ethernet cables

For the ethernet cables used in this project, we have chosen the URGREEN Cat6 Ethernet Cable. The major difference between URGREEN Cat6 Ethernet Cable with other different brands we can compare is **D-Link Cat6 Ethernet Cable**.

Specification	D-Link Cat6 Ethernet Cable	URGREEN Cat6 Ethernet Cable
Data Speed	Up to 1000 Mbps	Up to 1000 Mbps
Bandwidth	250 MHz	250 MHz
Connectors	Standard RJ45 connectors (may not always be gold-plated)	Gold-plated RJ45 connectors for corrosion resistance
Durability	Similar durable PVC outer jacket	Flexible PVC outer jacket for protection and long-lasting use
Additional Features	Standard performance for general-purpose networking	Enhanced stability for HD streaming and cloud computing
Price	20.00 for 2 meters	9.85 for 2 meters

Both the URGREEN and D-Link Cat6 Ethernet Cables have a bandwidth of 250 MHz and support data rates of up to 1000 Mbps. The D-Link cable is designed for general-purpose networking, featuring standard RJ45 connectors (which may not always be gold-plated), a durable PVC outer jacket, and reliable performance, making it a practical choice for everyday use. The URGREEN cable, on the other hand, is suitable for HD streaming and cloud computing, with gold-plated RJ45 connections that prevent corrosion, a flexible PVC outer jacket that increases longevity, and better stability for demanding applications. Finally, since the URGREEN Cat6 Ethernet Cable costs 9.85 for 2 meters, which is half as much as the D-Link cable, it is a more affordable choice.

5) Wireless access point

For the wireless access point used in this project, we have chosen the Tp-Link AX3600-EAP660 HD. The other different brands that we can compare is **Ubiquiti UniFi U6-LR** since both utilise the same Wi-Fi standard.

Specification	Ubiquiti UniFi U6-LR	Tp-Link AX3600-EAP660 HD
Wi-Fi Standard	Wi-Fi 6 (802.11ax)	Wi-Fi 6 (802.11ax)
Max Throughput	3 Gbps (2.4 GHz: 600 Mbps, 5 GHz: 2.4 Gbps)	3.6 Gbps (2.4 GHz: 1.1 Gbps, 5 GHz: 2.4 Gbps)
Ethernet Port	1 × 1GbE	1 × 2.5GbE
Antennas	4 internal antennas	8 internal antennas
Range	Long-range coverage (ideal for large areas)	Moderate (focus on performance)
Advanced Features	MU-MIMO, Beamforming	MU-MIMO, Beamforming, Airtime Fairness
Price (RM)	1758.00	1153.00

Both wireless access point shares the same Wi-Fi standard, which is 802.11ax with latest Wi-Fi 6 quality features. However, the max throughput of Tp-Link AX3600-EAP660 HD is higher compared to Ubiquiti UniFi U6-LR, with 3.6Gbps and 3Gbps respectively. Tp-Link AX3600-EAP660 HD also has more ethernet ports which allow for link aggregation. This feature is highly significant especially in a high-density environment to increase bandwidth and throughput. Besides, Tp-Link AX3600-EAP660 HD has 4 more internal antennas, thus extending omni-directional coverage. Since the building is about medium size (40mx27m), larger coverage area is less needed compared to higher bandwidth performance, so Tp-Link AX3600-EAP660 HD is much preferred here. It also introduces new feature (Airtime Fairness) where faster client will automatically receive more bandwidth so that their individual connection speed won't be bogged down by slower clients. The only downside here is the price where Tp-Link AX3600-EAP660 HD is more expensive than Ubiquiti UniFi U6-LR, but this can be justified with its greater ethernet port numbers, antenna quantity and the extra advanced features.

6) Fiber optics

For the fiber optics used in this project, we have chosen the FS OM4-LC/UPC-LC/UPC-DX-1M Fiber Optic. The other brand model we can compare is the **Fibertronics OS2-LC/UPC-LC/UPC Duplex-3M Fiber Optic**.

Specification	OS2-LC/UPC-LC/UPC Duplex-3M Fiber Optic	OM4-LC/UPC-LC/UPC-DX-1M Fiber Optic
Brand	Fibertronics	FS
Fiber Mode	OS2 Singlemode 9/125 μ m	OM4 Multimode 50/125 μ m
Type	UC LPC Duplex	UC LPC Duplex
Jacket Type	PVC	OFNR
Electrical Interference Immunity	Yes	Yes
Pull-proof	Yes	Yes
Price (RM)	14.6/m	20/m

First, the biggest difference between these two fiber optics is singlemode fiber and multimode fiber. Singlemode is typically ideal for long distance while multimode for short distance. Besides, singlemode has a higher bandwidth while multimode has a higher data rate. However, we chose multimode fiber optics for our building as it meets the specific requirements of our medium size building. For the bandwidth aspect, we could improve its performance with the assistance of routers and wireless access point. On the other hand, they each have different jacket type. OFNR is much more fire resistant and non-conductive, thus potentially lowering maintenance costs associated with fire hazards and damage repairs. Although FS OM4-LC/UPC-LC/UPC-DX-1M Fiber Optic has its downsides from performance and price aspect, we still choose it due to it providing higher safety measures to accommodate the building for the next 20 years.

3.4 Reflection of Task 3

In this task, we gained a deeper understanding of network devices and functions in constructing a building with a high-performance network. We understand the specifications, purposes, and contributions to efficient network architecture such as routers, switches, patch panels, wireless access points, ethernet cables, and fibre optics.

The main things are we learn how to select networking devices based on the provided budget (RM1 million). Budget constraints required us to compare different brands and models to find the most economical options while maintaining high performance and reliability. This experience taught me the importance of balancing cost-effectiveness with technological requirements. For example, I analysed differences in performance, security, and scalability between products like TP-Link routers and Asus routers. These comparisons highlighted the factors like brand reputation and material quality impact the price and suitability of each device for the project.

Overall, this task enhanced my understanding of network design, equipment selection, and budget management.

Task 4: Identification of Floor Cable Connections

4.1 Identify the work areas on floor plan



Figure 4.1.1: Work Area at Ground Floor

Work Area 1 shown in Figure 4.1.1 is a student meeting room. It is eight meters wide and nine meters long. It is perfect for cooperative group discussions and gatherings because it has an oval table with six chairs around it. A workstation and a projector are among the useful equipment in the room that supports student activities and helps to improve group project discussions.

Work Area 2 shown in Figure 4.1.1 is student lounge measures 10 meters in width and 14 meters in length. Students can use this lounge to relax or study before class. It is equipped with eight workstations for academic tasks, along with a rest area featuring three comfortable sofas for relaxation. Along with a printer to let students print study materials, the area also has a selection of books for study purposes. Furthermore, a Wi-Fi access point is set up to offer dependable and simple network connectivity, ensuring that students stay connected and productive while using the student lounge.

Work Area 3 and 4 as shown in Figure 4.1.1 are both general labs spanning 14 meters in length and 10 meters in width. These general labs are intended for student use during classes. Each lab is equipped with a total of 60 PC workstations providing students hands-on learning and collaborative projects.

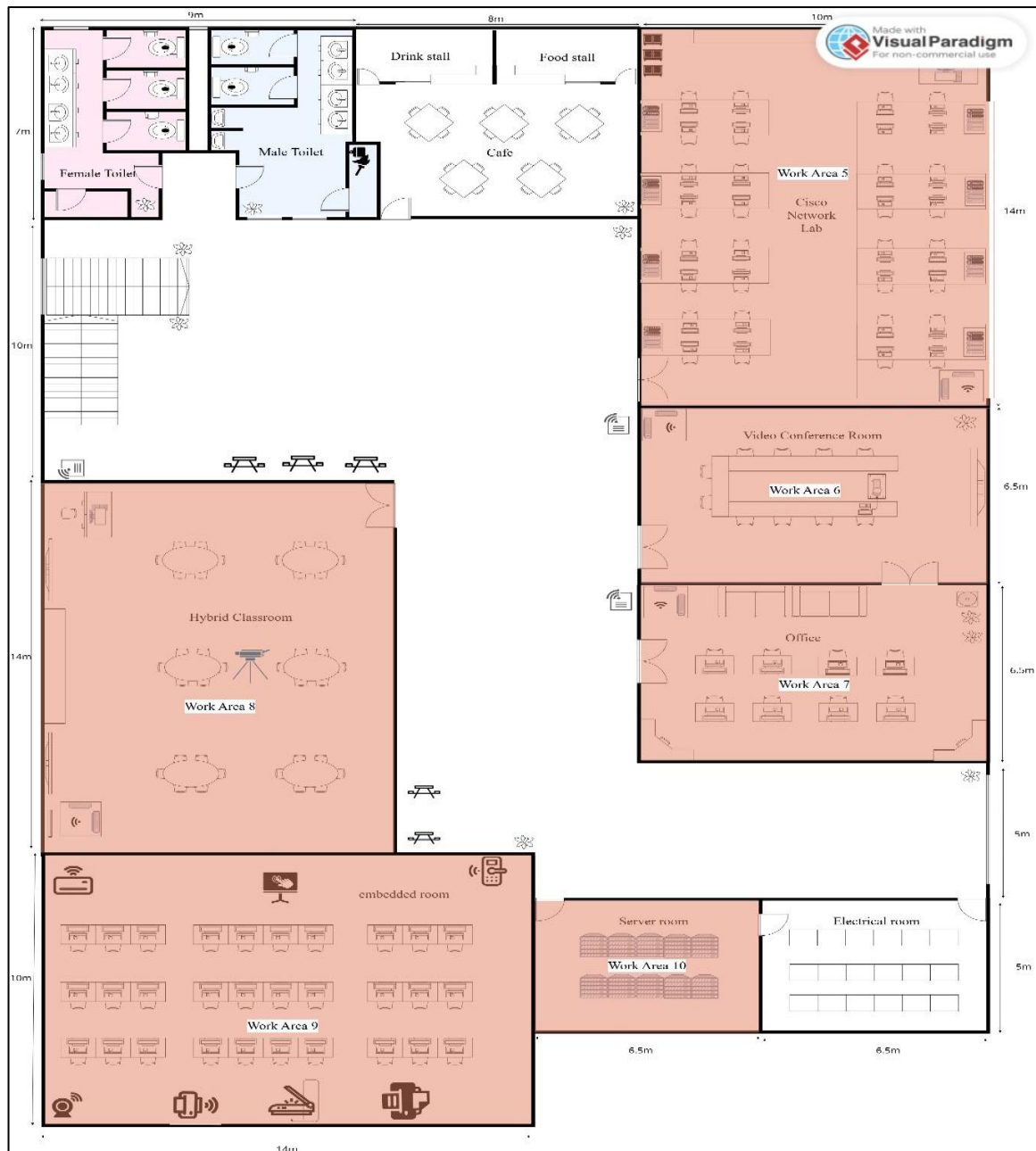


Figure 4.1.2: Work Area at First Floor

The Cisco Network Lab for work area 5 shown in Figure 4.1.2 is 10 meters wide and 14 meters long. It is in the upper-right corner of the first level. There are four rows of tables with four computer workstations each on the left and right sides of the lab, providing students lots of space for learning. There is also an instructor's desk at the top-right corner, equipped with a workstation. Additionally, a Wi-Fi access point is installed in the bottom-right corner for wireless connectivity.

Video Conference Room is located at Work Area 6 shown in Figure 4.1.2 which designed for meetings and virtual collaboration. The room measures 6.5 meters in length 10m in width and is organized around a large central conference table with 8 chairs. At end of the

table, there is a designated workstation equipped with a computer and projector. The room also includes a large display screen on the wall collaborate with projector to facilitate presentations and virtual meetings. Additionally, a Wi-Fi access point is installed on the left side of the room ensure high speed Wi-Fi connectivity for video conferencing.

For the Work Area 7 shown in Figure 4.1.2, it is a lecturer's office with dimensions of 6.5 meters in length and 10 meters in width. The office contains eight workstations, arranged in two rows of four, each equipped with a chair and computer setup. There are two sofas positioned at the top of the room, providing a comfortable seating area. Additionally, a Wi-Fi access point is installed in the top-left corner to ensure wireless connectivity.

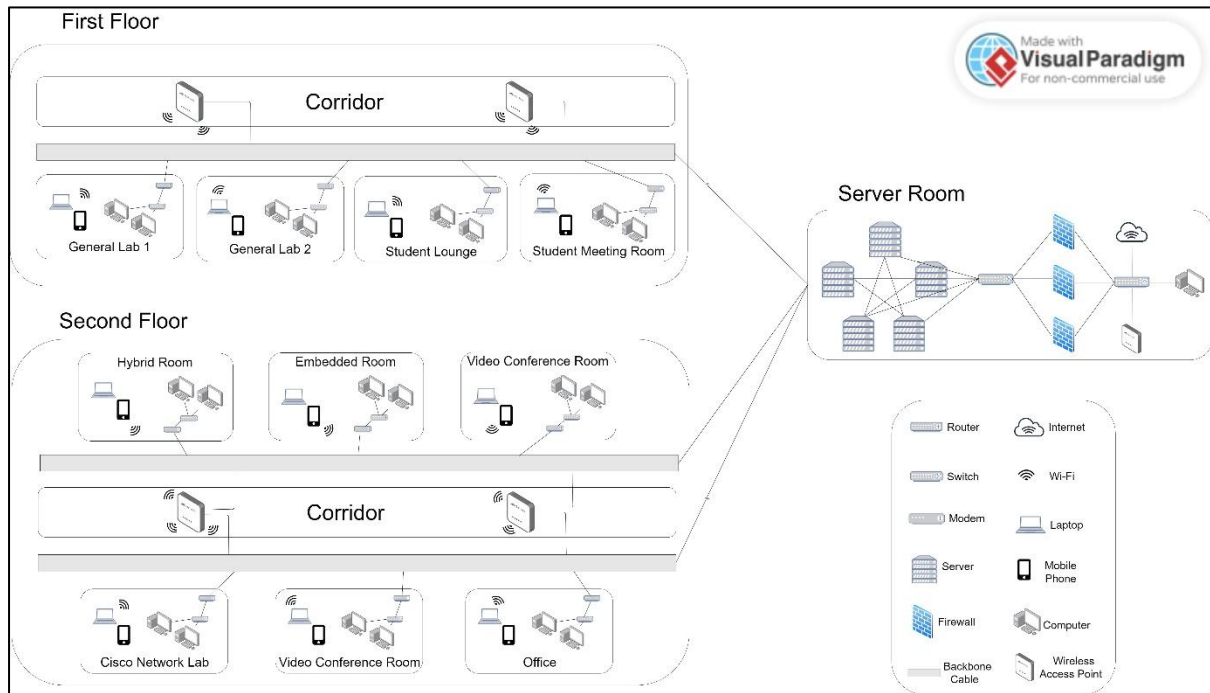
Work Area 8 shown in Figure 4.1.2 is hybrid classroom. The hybrid classroom must be setup for best performance and efficiency for teaching and learning. A video conferencing camera set up in the middle of the room allows for distant participation and hybrid instruction. In addition, the classroom has a teacher's workstation, a presentation display screen, and a Wi-Fi access point for constant online connectivity.

Work Area 9 shown in Figure 4.1.2 is embedded lab designed to facilitate learning in embedded systems and IoT. It has 30 workstations, ensuring that students have the resources they need to complete practical tasks. The lab contains a variety of IoT equipment, such as digital sensors (temperature, motion, and pressure sensors), Wi-Fi, and Bluetooth communication devices, which allow for real-time data collection and wireless networking. IoT is supported by additional peripherals like as webcams, printers, and scanners.

Work Area 10 shown in Figure 4.1.3 is server room. The room measures 6.5 meters in width and 5 meters in length, and it is equipped with arranged racks that have servers, network switches, routers, and other necessary hardware. The server room is designed to allow high-capacity data processing, storage and connection making it the foundation of IT operations.

4.2 The distribution and connection of devices

4.2.1 Network Distribution Diagram



4.2.1.1 Network Distribution Diagram

Figure 4.2.1.1 shows the entirety of the network distribution within the faculty of computing. This network distribution diagram aims to visually depict the layout and connectivity of the devices in each work area, including routers, switches, wireless access points and end devices. This diagram will also show the cabling routes for horizontal and vertical connections, making it clear what the logical and physical topology of the network is.

Figure 4.2.1.2 and Figure 4.2.1.3 shows the network distribution of the work areas of the first floor, including two general labs, student lounge and student meeting room. The workstations are connected to the router and then to the modem through ethernet cable. This is to ensure seamless connectivity and better network performance as compared to wireless connection. Lastly the modem is connected through optic fibre to the wireless access point. The end devices such as mobile devices and laptop are connected wirelessly to the wireless access point which provides the Internet IP addresses.

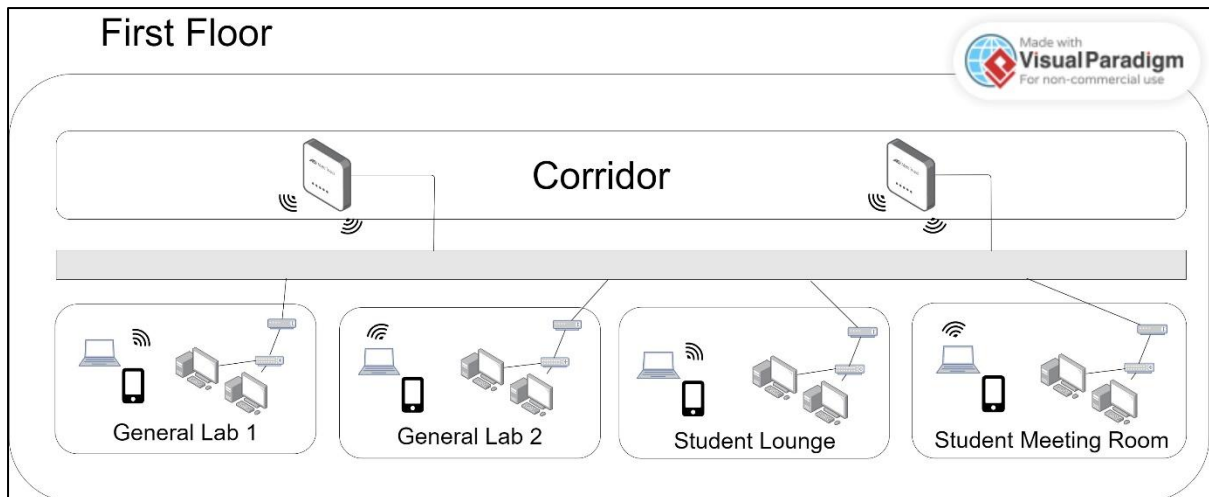


Figure 4.2.1.2 Network Distribution Diagram – First Floor

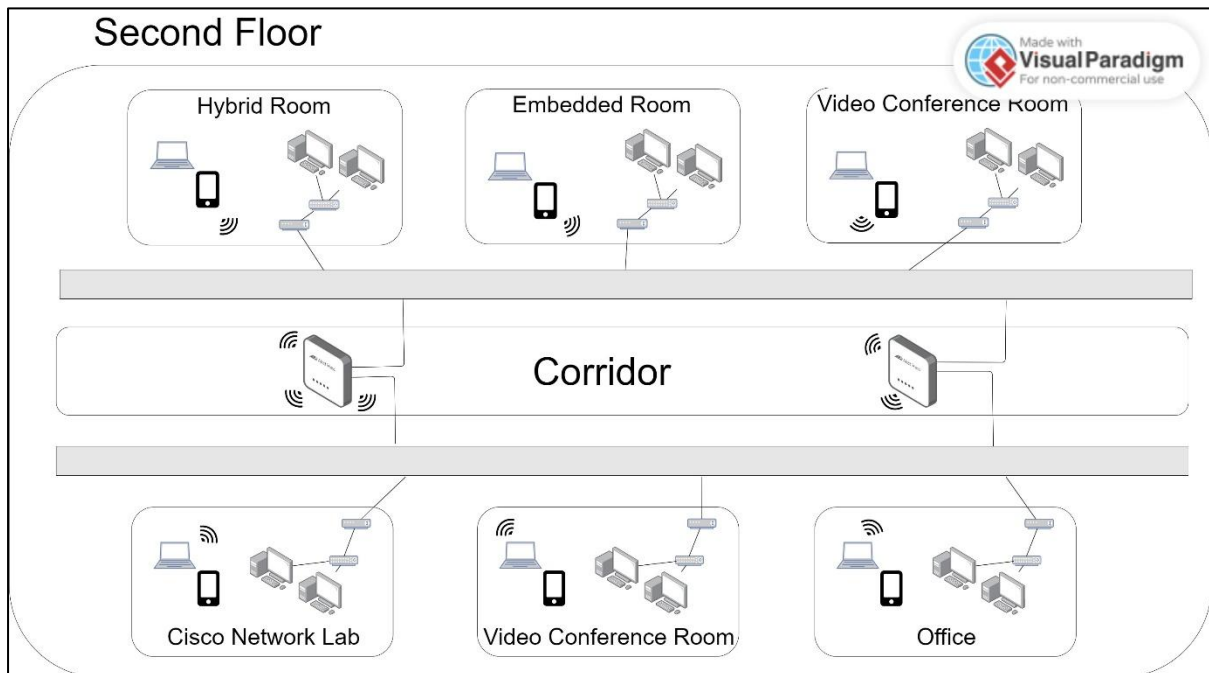


Figure 4.2.1.3 Network Distribution Diagram – Second Floor

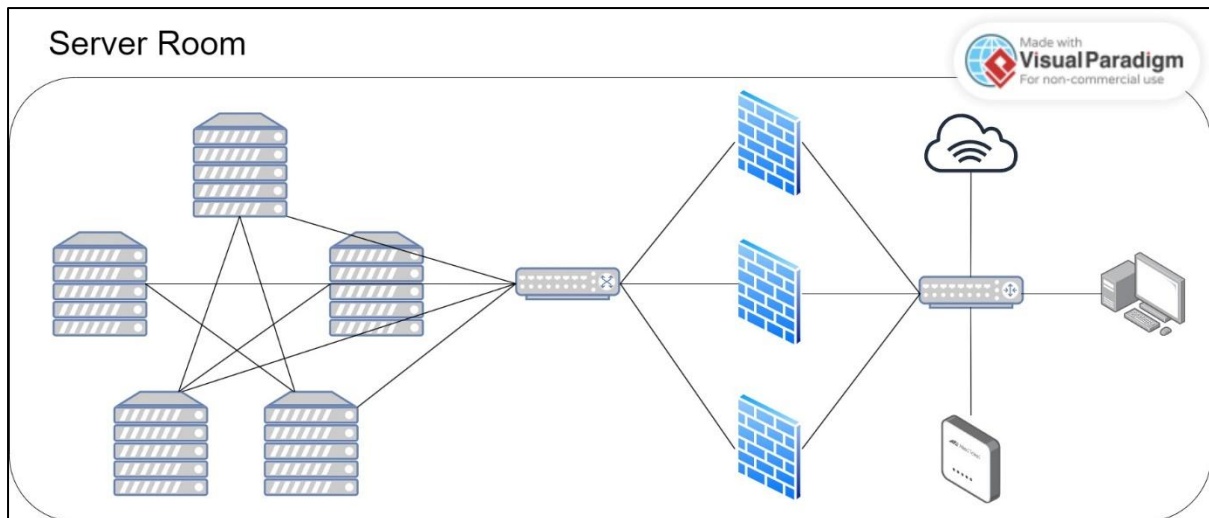


Figure 4.2.1.4 Network Distribution Diagram – Server Room

As seen in Figure 4.2.1.4, the servers are interconnected using a mesh topology. This network distribution can increase reliability and longevity of the server's services as the other server is still able to connect to the Internet even when one server is malfunction or under reparation. This design also promotes scalability as new server can be added without interrupting the existing network, making it better for future growth.

The faculty of computing's servers are mainly used as a centralized storage for research data, student work, course materials and faculty resources. Besides, it acts as a host core network infrastructure to manage traffic between the internal network and the Internet.

In order to maintain network security, the servers are directed to the firewalls through switches before entering the internal network infrastructures through the router. Cisco networking lab, embedded lab, hybrid lab and general labs are highly dependent on the servers as they deliver quality networking with low latency to ensure seamless connection data system management.

4.2.2 Room Level Network Distribution Diagram

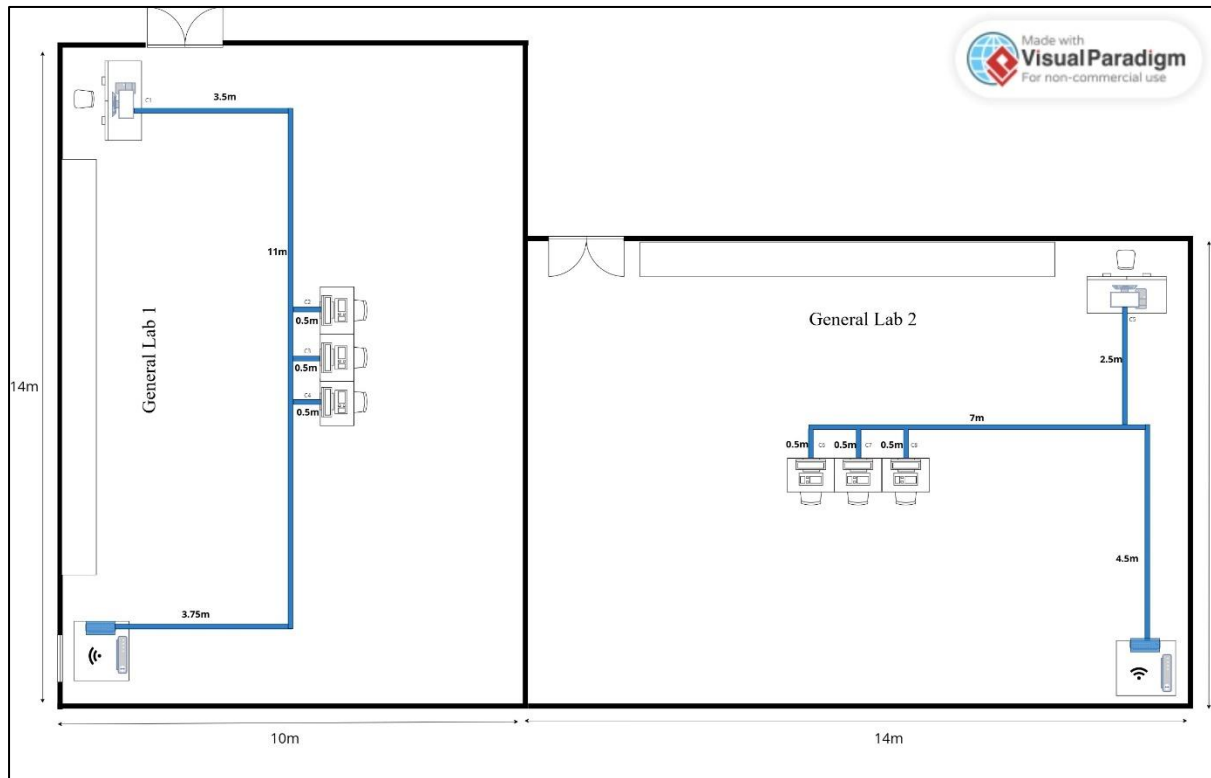


Figure 4.2.2.1: General Lab

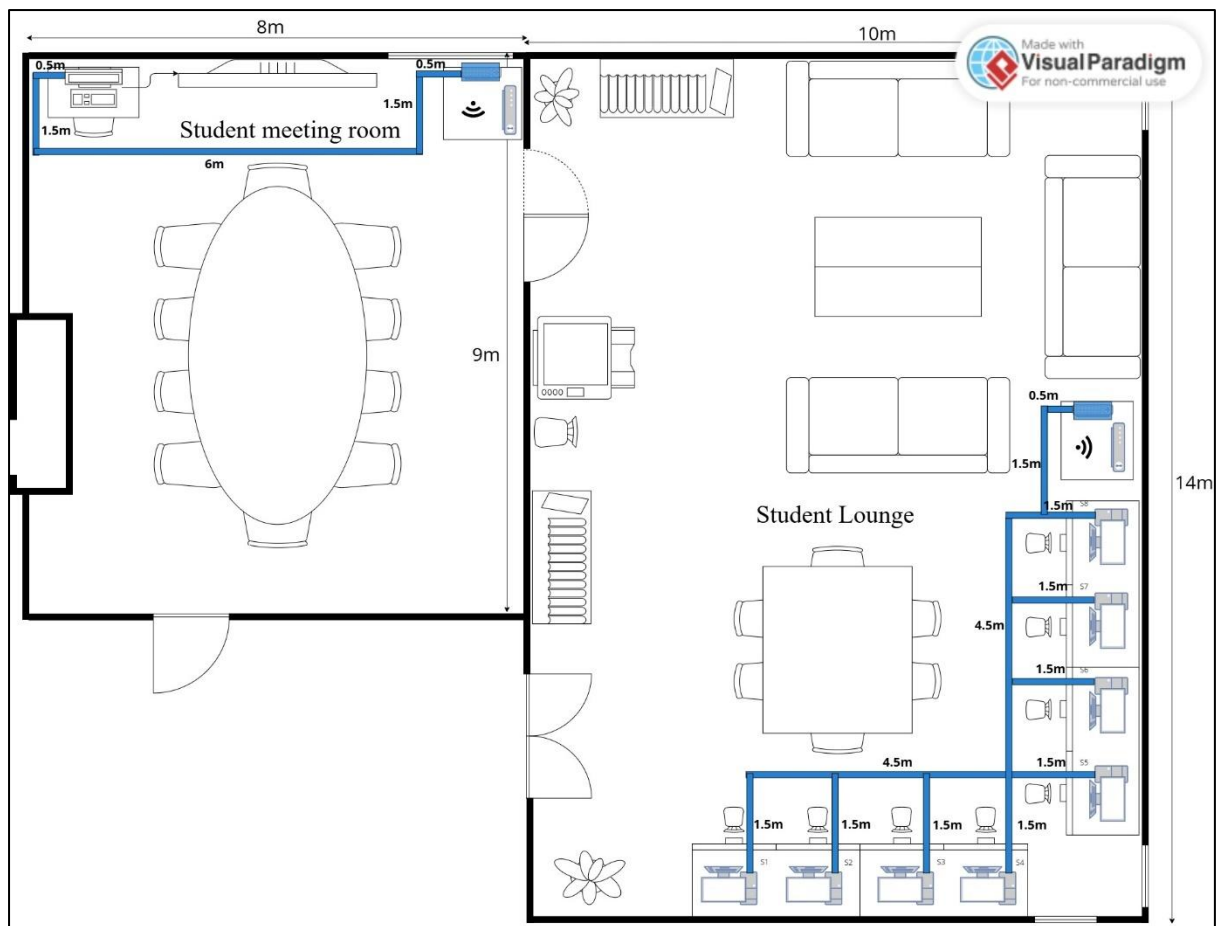


Figure 4.2.2.2: Student Meeting Room and Student Lounge

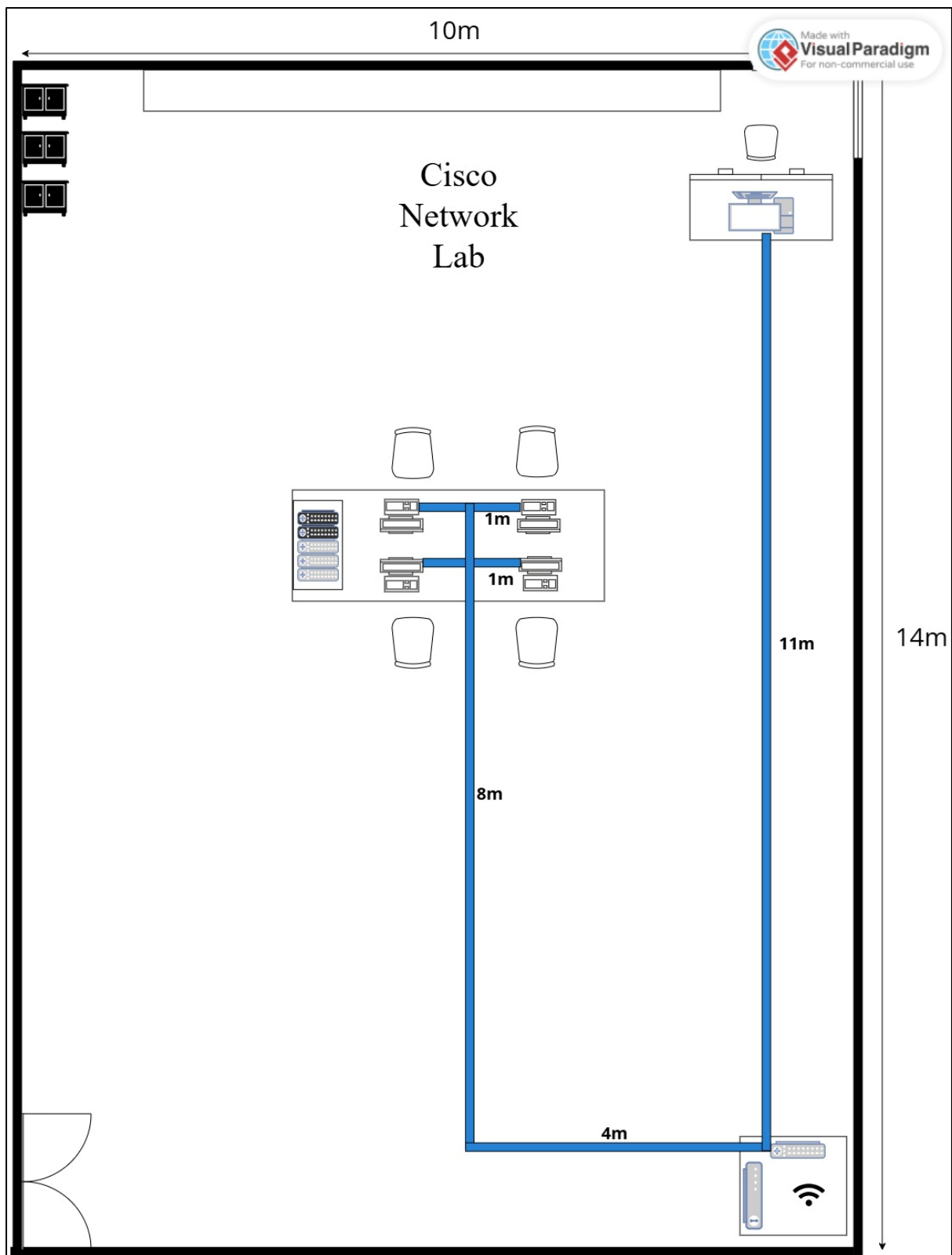


Figure 4.2.2.3: Cisco Network Lab

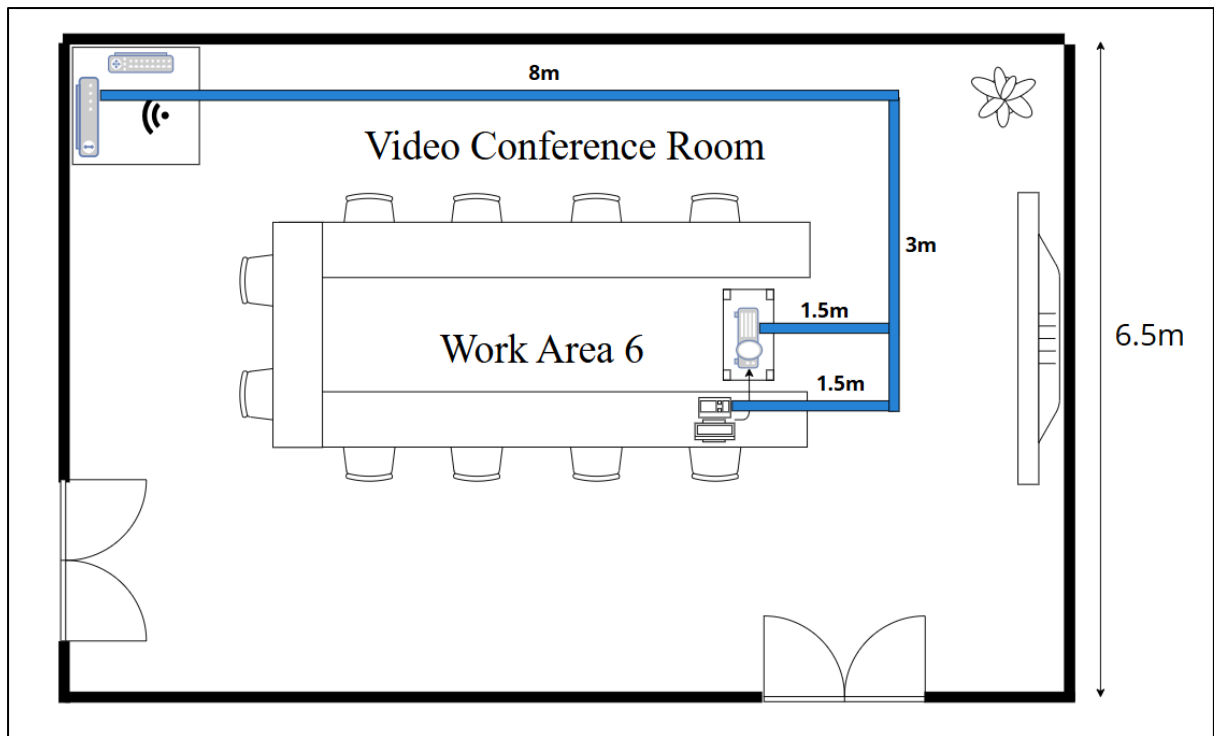


Figure 4.2.2.4: Video Conference Room

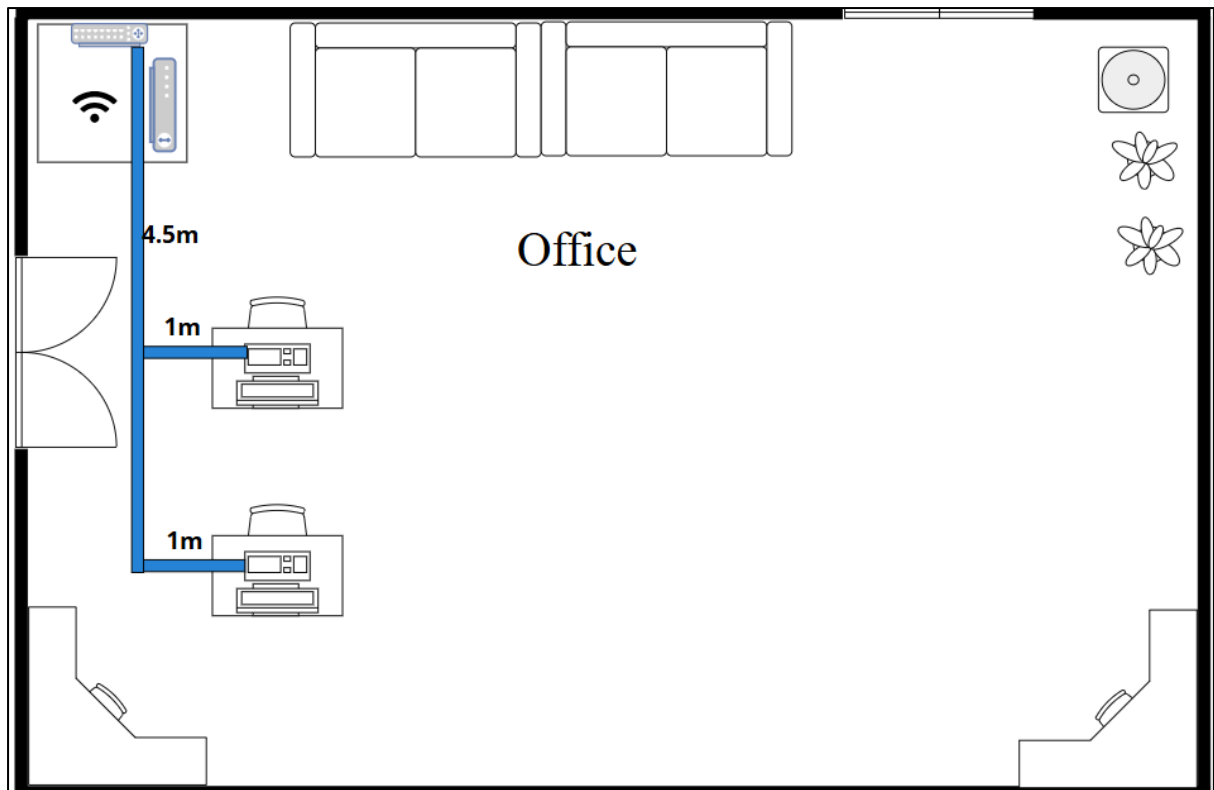


Figure 4.2.2.5: Office

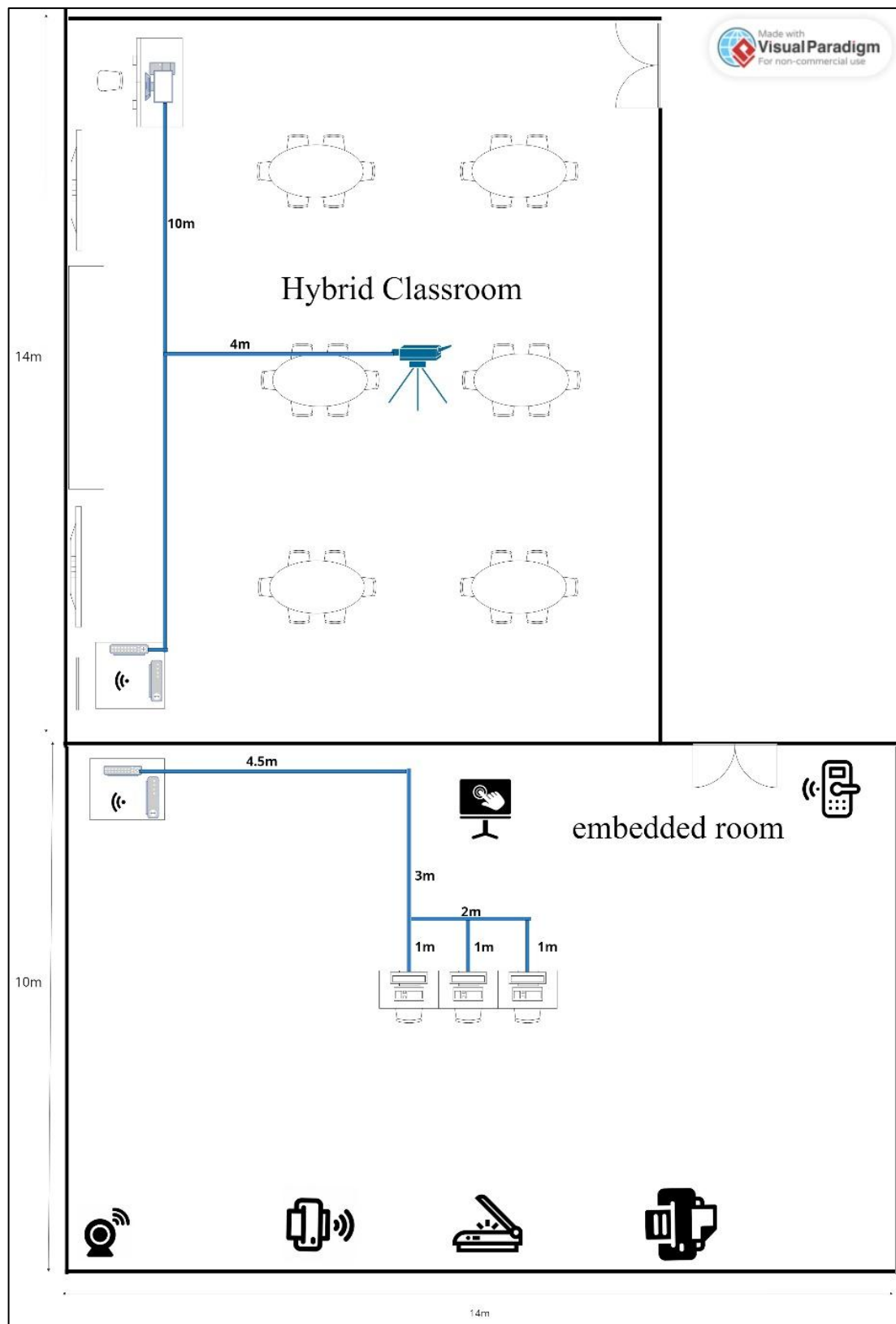


Figure 4.2.2.6: Hybrid Classroom and Embedded Room

4.2.3 Floor Diagram

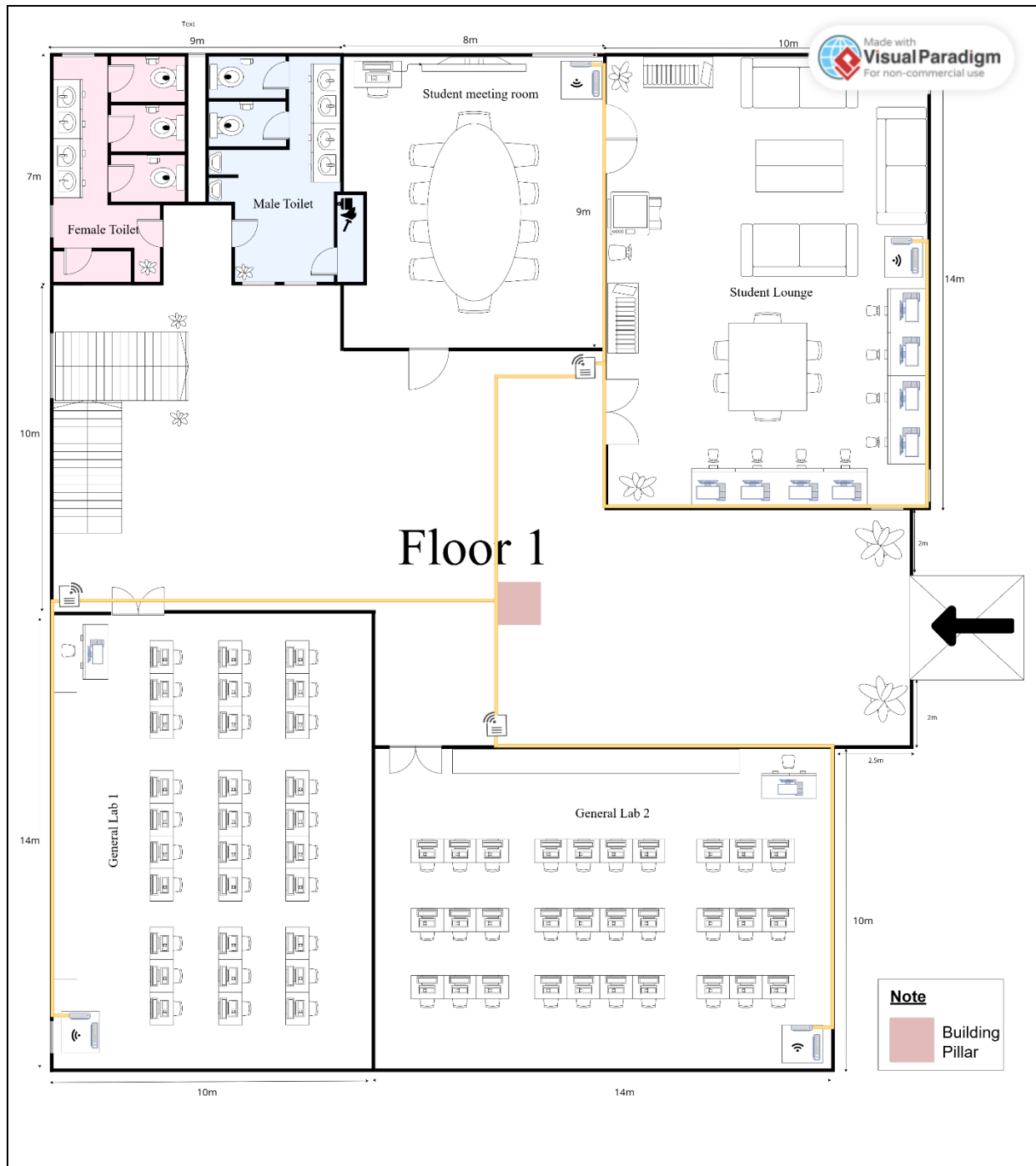


Figure 4.2.3.1 Floor Diagram – First Floor

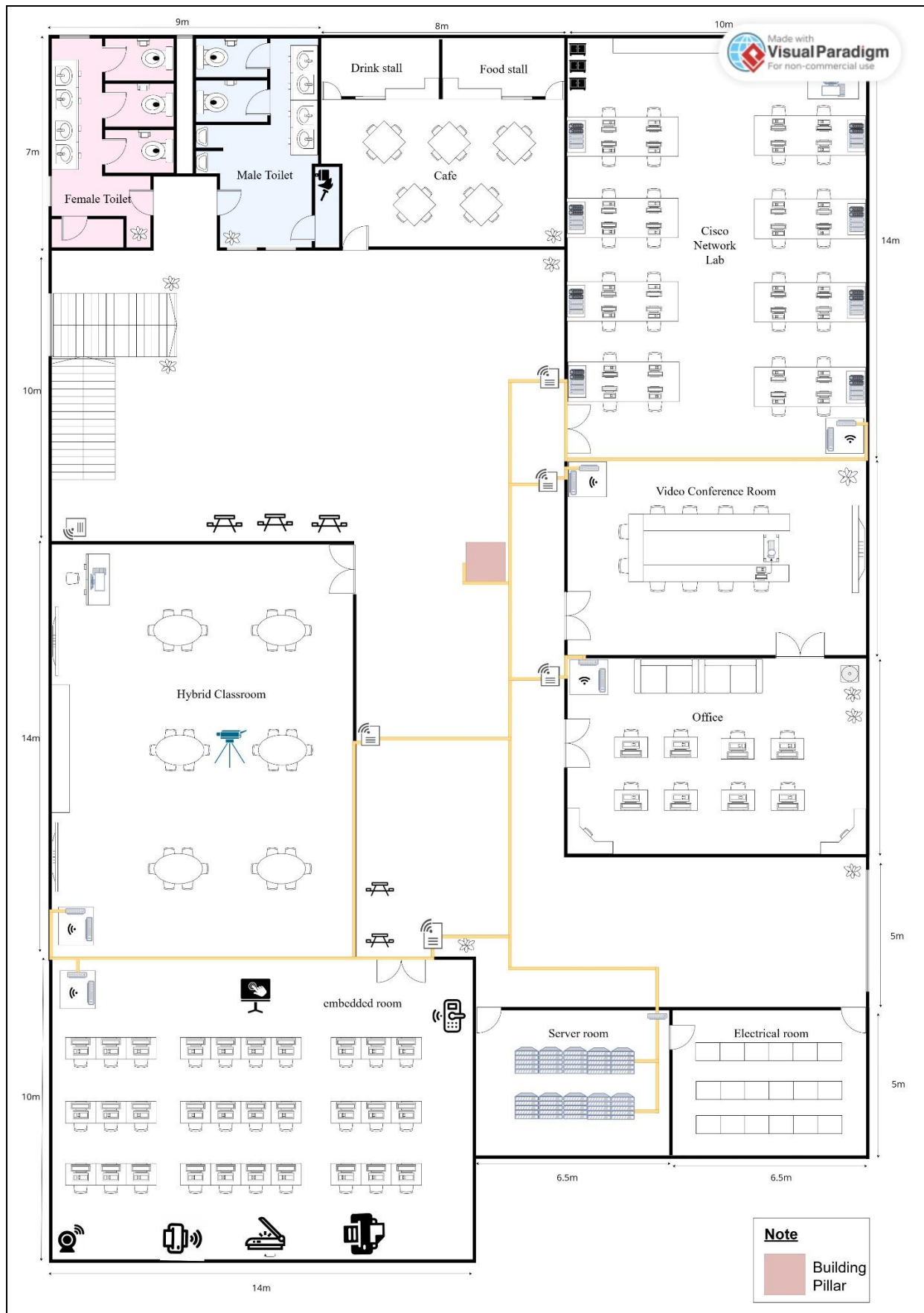


Figure 4.2.3.2 Floor Diagram – Second Floor

4.3 Number of connections, patch cords and switch ports needed

Work Area	Number of Devices	Type of Devices	Number of Ports Required
Student Meeting Room	6 Workstations	PCs, Wi-Fi Access Point	7
Student Lounge	8 Workstations, 1 Printer	PCs, Printer, Wi-Fi	10
General Labs (x2)	60 Workstations	PCs, Instructor's Workstation	62
Cisco Network Lab	16 Workstations, 1 Instructor's Workstation	PCs, Wi-Fi Access Point	18
Video Conference Room	1 Workstation, 1 Projector	PC, Projector, Wi-Fi	3
Lecturer's Office	8 Workstations	PCs, Wi-Fi Access Point	9
Hybrid Classroom	1 Teacher Workstation, Video Conferencing Camera	Teacher PC, Camera, Wi-Fi	3
Embedded Lab	30 Workstations	PCs, IoT Devices, Wi-Fi	31
Server Room	Multiple Devices	Servers, Switches, Routers	10
Total			153

Table 4.3.1: Number of Switch Ports Needed

Work Area	Number of Devices	Type of Devices	Number of Ports Required
Student Meeting Room	6 Workstations	PCs, Wi-Fi Access Point	7
Student Lounge	8 Workstations, 1 Printer	PCs, Printer, Wi-Fi	10
General Labs (x2)	60 Workstations	PCs, Instructor's Workstation	62
Cisco Network Lab	16 Workstations, 1 Instructor's Workstation	PCs, Wi-Fi Access Point	18
Video Conference Room	1 Workstation, 1 Projector	PC, Projector, Wi-Fi	3
Lecturer's Office	8 Workstations	PCs, Wi-Fi Access Point	9
Hybrid Classroom	1 Teacher Workstation, Video Conferencing Camera	Teacher PC, Camera, Wi-Fi	3

Embedded Lab	30 Workstations	PCs, IoT Devices, Wi-Fi	31
Server Room	Multiple Devices	Servers, Switches, Routers	10
Total			153

Table 4.3.2: Number of Patch Cords Needed

Work Area	Number of Devices	Type of Devices	Number of Ports Required
Student Meeting Room	6 Workstations	PCs, Wi-Fi Access Point	14
Student Lounge	8 Workstations, 1 Printer	PCs, Printer, Wi-Fi	20
General Labs (x2)	60 Workstations	PCs, Instructor's Workstation	124
Cisco Network Lab	16 Workstations, 1 Instructor's Workstation	PCs, Wi-Fi Access Point	36
Video Conference Room	1 Workstation, 1 Projector	PC, Projector, Wi-Fi	6
Lecturer's Office	8 Workstations	PCs, Wi-Fi Access Point	18
Hybrid Classroom	1 Teacher Workstation, Video Conferencing Camera	Teacher PC, Camera, Wi-Fi	6
Embedded Lab	30 Workstations	PCs, IoT Devices, Wi-Fi	62
Server Room	Multiple Devices	Servers, Switches, Routers	20
Total			306

Table 4.3.3: Number of RJ45 Connector Needed

4.4 Identify cable types and length

Based on the floor diagram from task 1, the cable length can be measured almost accurately. Each floor in the building has an approximate height of 4m.

Area	Length(m)	Total Price of Cable Usage
Ground Floor		
Room		

Student meeting room	10	200
Student lounge	23	460
General lab 1	54.5	1090
General lab 2	54.5	1090
Ground Floor Room Total		2840
First Floor		
Room		
Cisco Network Lab	62	1240
Video Conference Room	9	180
Lecturer office	17.5	350
Hybrid classroom	14.5	290
Embedded lab	53	1060
First Floor Room Total		3120

First Floor		
Room		
Path	Length(m)	Price of cables used(RM)
Server room to General Lab 1	48	960
Server room to General Lab 2	47	940
Server room to Student Lounge	52	1040
Server room to Student Meeting Room	42	840
Ground Floor Room Total	189	3780
Total – First Floor		
Total length of wire to connect the room	189	3780
Total length of wire in different type	2840	56800
Total length of wire needed	3029	60580
Second Floor		
Server Room to Cisco Network Lab	38	760
Server Room to Video Conference Room	23	460

Server Room to Lecturer Office	17	340
Server Room to Hybrid classroom	39	780
Server Room to Embedded Lab	22	440
Second Floor Room Total	139	2780
Total - Second Floor		
Total length of wire to connect the room(m)	139	2780
Total length of wire in different type(m)	3120	62400
Total length of wire needed	3259	65180
Total (First Floor + Second Floor)	6288	125760

Explanation:

We detailed the distribution strategy and device connections. To enhance connectivity, selecting the appropriate cable types is essential. After thorough discussion, our team decided on a combination of CAT6 twisted pair cables and fiber optic cables for the building's network infrastructure.

CAT6 twisted pair cables will be used for both horizontal and vertical cabling in areas such as labs and lecture halls. It was because CAT6 can provide data transmission of as high as 10 Gbps due to the 550 MHz bandwidth. Moreover, the thicker copper wire allows for better dissipation of heat, which will enable the cable to perform under PoE, decreasing cabling complexity and cost. CAT6 cables are also the perfect fit for Ethernet applications within the building. We chose UTP CAT6 cables because they can reduce crosstalk and minimize electromagnetic interference from either natural or artificial sources.

Routers in the server room and connecting to an ISP will be connected using fiber optic cables. These cables use light to transmit data, enabling high transfer rates without degradation of the signal. The cables have a bandwidth of 10 Gbps, hence are suitable for handling massive volumes of data and maintaining quality signals over long distances.

In conclusion, the estimated total cable length of both first and second floor is 6288m. The estimated cable length will include both CAT6 and fiber optic cables that will ensure seamless internal connectivity between devices and a strong, reliable connection to the ISP.

4.5 Reflection of Task 4

In this task we identified the work areas in the faculty building as well as create cable layout diagrams for each floor and individual work areas. The cable first connects from end devices to switches, which then connects to routers, wireless access point and finally to the server room, which serves as the core network infrastructure of the faculty building.

Next, we identified the number of switch ports, patch cords and RJ45 connectors for each work areas. Additionally, the estimate total cable length is calculated by referring to the floor diagrams. These calculations allow us to estimate the total budget required for the cabling wires, thus allowing us to review the economic feasibility of the network infrastructure.

In conclusion, the estimated total cable length of both first and second floor is 6288m, summing up to a total price of RM125,760. The estimated cable length will include both CAT6 and fiber optic cables to ensure fast connectivity.

Task 5: Subnet

5.1 Get the Network Address from your lecturer

Group/Section	Network Address
1	10.16.0.0/12
2	10.32.0.0/12
3	10.48.0.0/12
4	10.64.0.0/12
5	10.80.0.0/12
6	10.96.0.0/12
7	10.112.0.0/12
8	10.128.0.0/12
9	10.144.0.0/12
10	10.176.0.0/12

Table 5.1.1

Table 5.1.1 shows the list of network addresses of each group. The network address of our group is 10.128.0.0/12.

Network address details:

IP address (decimal)	10.128.0.0
IP address (binary)	00001010 10000000 00000000 00000000
Subnet Mask(decimal)	255.240.0.0
Subnet Mask(binary)	11111111 11110000 00000000 00000000

We partitioned the subnet mask into a network portion (12 bits) and a host portion (20 bits). By performing the AND operation of the IP address with the subnet mask, we identified the network address as 10.128.0.0 and the broadcast address as 10.143.255.255. The valid address range within our building spans from 10.128.0.1 to 10.143.255.254.

Network Address	10.128.0.0
Broadcast Address	10.143.255.255
Range of usable address	10.128.0.1 - 10.143.255.254

5.2 Divide it in the best possible way for your network

5.2.1 Network distribution for different work area

In order to improve internet speed and enhance security of the faculty work areas, subnetting is done to divide the IP address space into fixed-size classes. Based on the network design requirements, we identified that we need 8 subnets to accommodate 10 different work areas, where some subnets are used to manage several work areas. To achieve these, 3 extra bits ($2^3=8$) is borrowed from host portion to construct subnet IP address. The table below listed each subnet network portion and host portion and the distribution of subnet for each area.

Subnet	Network portion	Subnet bits	Host portion		IP address
0	00001010 1000	000	0 0000 0000 0000 0000	Network Address	10.128.0.0
			1 1111 1111 1111 1111	Broadcast Address	10.129.255.255
1	00001010 1000	001	0 0000 0000 0000 0000	Network Address	10.130.0.0
			1 1111 1111 1111 1111	Broadcast Address	10.131.255.255
2	00001010 1000	010	0 0000 0000 0000 0000	Network Address	10.132.0.0
			1 1111 1111 1111 1111	Broadcast Address	10.133.255.255
3	00001010 1000	011	0 0000 0000 0000 0000	Network Address	10.134.0.0
			1 1111 1111 1111 1111	Broadcast Address	10.135.255.255

4	00001010 1000	100	0 0000 0000 0000 0000	Network Address	10.136.0.0
			1 1111 1111 1111 1111	Broadcast Address	10.137.255.255
5	00001010 1000	101	0 0000 0000 0000 0000	Network Address	10.138.0.0
			1 1111 1111 1111 1111	Broadcast Address	10.139.255.255
6	00001010 1000	110	0 0000 0000 0000 0000	Network Address	10.140.0.0
			1 1111 1111 1111 1111	Broadcast Address	10.141.255.255
7	00001010 1000	111	0 0000 0000 0000 0000	Network Address	10.142.0.0
			1 1111 1111 1111 1111	Broadcast Address	10.143.255.255

Distribution of subnet for different work area:

Subnet	Work Area	Network Address	Broadcast Address	Range of usable address	Subnet Mask Address
0	Server Room	10.128.0.0	10.129.255.255	10.128.0.1 - 10.129.255.254	/15 255.254.0.0
First Floor					
1	General Lab 1	10.130.0.0	10.131.255.255	10.130.0.1 - 10.131.255.254	/15 255.254.0.0

2	General Lab 2	10.132.0.0	10.133.255.255	10.132.0.1 - 10.133.255.254	/15 255.254.0.0
3	Student Lounge and Student Meeting Room	10.134.0.0	10.135.255.255	10.134.0.1 - 10.135.255.254	/15 255.254.0.0
Second Floor					
4	Office & Video Conference Room	10.136.0.0	10.137.255.255	10.136.0.1 - 10.137.255.254	/15 255.254.0.0
5	Cisco Network Lab	10.138.0.0	10.139.255.255	10.138.0.1 - 10.139.255.254	/15 255.254.0.0
6	Embedded Room	10.140.0.0	10.141.255.255	10.140.0.1 - 10.141.255.254	/15 255.254.0.0
7	Hybrid Classroom	10.142.0.0	10.143.255.255	10.142.0.1 - 10.143.255.254	/15 255.254.0.0

5.2.2 Network distribution for workstation in different work areas

First Floor:

1. General Lab 1

Devices	IP Address
PC1	10.130.0.1
PC2	10.130.0.2
PC3	10.130.0.3
PC4	10.130.0.4
PC5	10.130.0.5
PC6	10.130.0.6
PC7	10.130.0.7

PC8	10.130.0.8
PC9	10.130.0.9
PC10	10.130.0.10
PC11	10.130.0.11
PC12	10.130.0.12
PC13	10.130.0.13
PC14	10.130.0.14
PC15	10.130.0.15
PC16	10.130.0.16
PC17	10.130.0.17
PC18	10.130.0.18
PC19	10.130.0.19
PC20	10.130.0.20
PC21	10.130.0.21
PC22	10.130.0.22
PC23	10.130.0.23
PC24	10.130.0.24
PC25	10.130.0.25
PC26	10.130.0.26
PC27	10.130.0.27
PC28	10.130.0.28
PC29	10.130.0.29
PC30	10.130.0.30
PC31	10.130.0.31
PC32	10.130.0.32
PJ1	10.130.0.33
WAP1	10.130.0.34

2. General Lab 2

Devices	IP Address
PC33	10.132.0.1
PC34	10.132.0.2
PC35	10.132.0.3
PC36	10.132.0.4
PC37	10.132.0.5
PC38	10.132.0.6
PC39	10.132.0.7
PC40	10.132.0.8
PC41	10.132.0.9
PC42	10.132.0.10
PC43	10.132.0.11
PC44	10.132.0.12
PC45	10.132.0.13

PC46	10.132.0.14
PC47	10.132.0.15
PC48	10.132.0.16
PC49	10.132.0.17
PC50	10.132.0.18
PC51	10.132.0.19
PC52	10.132.0.20
PC53	10.132.0.21
PC54	10.132.0.22
PC55	10.132.0.23
PC56	10.132.0.24
PC57	10.132.0.25
PC58	10.132.0.26
PC59	10.132.0.27
PC60	10.132.0.28
PC61	10.132.0.29
PC62	10.132.0.30
PC63	10.132.0.31
PC64	10.132.0.32
PJ2	10.132.0.33
WAP2	10.132.0.34

3. Student Lounge

Devices	IP Address
PC65	10.134.0.1
PC66	10.134.0.2
PC67	10.134.0.3
PC68	10.134.0.4
PC69	10.134.0.5
PC70	10.134.0.6
PC71	10.134.0.7
PC72	10.134.0.8
PRINTER1	10.134.0.9
WAP3	10.134.0.10

4. Student Meeting Room

Devices	IP Address
PJ4	10.135.0.1
MONITOR2	10.135.0.2
WAP4	10.135.0.3

Second floor:

5. Office

Devices	IP Address
PC73	10.136.0.1
PC74	10.136.0.2
PC75	10.136.0.3
PC76	10.136.0.4
PC77	10.136.0.5
PC78	10.136.0.6
PC79	10.136.0.7
PC80	10.136.0.8
WAP5	10.136.0.9

6. Video Conference Room

Devices	IP Address
PJ5	10.137.0.1
MONITOR	10.137.0.2
WAP6	10.137.0.3

7. Cisco Network Lab

Devices	IP Address
PC81	10.138.0.1
PC82	10.138.0.2
PC83	10.138.0.3
PC84	10.138.0.4
PC85	10.138.0.5
PC86	10.138.0.6
PC87	10.138.0.7
PC88	10.138.0.8
PC89	10.138.0.9
PC90	10.138.0.10
PC91	10.138.0.11
PC92	10.138.0.12
PC93	10.138.0.13
PC94	10.138.0.14
PC95	10.138.0.15
PC96	10.138.0.16
PC97	10.138.0.17
PC98	10.138.0.18
PC99	10.138.0.19
PC100	10.138.0.20
PC101	10.138.0.21
PC102	10.138.0.22

PC103	10.138.0.23
PC104	10.138.0.24
PC105	10.138.0.25
PC106	10.138.0.26
PC107	10.138.0.27
PC108	10.138.0.28
PC109	10.138.0.29
PC110	10.138.0.30
PC111	10.138.0.31
PC112	10.138.0.32
SWITCH1	10.138.0.33
SWITCH2	10.138.0.34
SWITCH3	10.138.0.35
SWITCH4	10.138.0.36
SWITCH5	10.138.0.37
SWITCH6	10.138.0.38
SWITCH7	10.138.0.39
SWITCH8	10.138.0.40
SWITCH9	10.138.0.41
SWITCH10	10.138.0.42
SWITCH11	10.138.0.43
SWITCH12	10.138.0.44
SWITCH13	10.138.0.45
SWITCH14	10.138.0.46
SWITCH15	10.138.0.47
SWITCH16	10.138.0.48
ROUTER1	10.138.0.49
ROUTER2	10.138.0.50
ROUTER3	10.138.0.51
ROUTER4	10.138.0.52
ROUTER5	10.138.0.53
ROUTER6	10.138.0.54
ROUTER7	10.138.0.55
ROUTER8	10.138.0.56
ROUTER9	10.138.0.57
ROUTER10	10.138.0.58
ROUTER11	10.138.0.59
ROUTER12	10.138.0.60
ROUTER13	10.138.0.61
ROUTER14	10.138.0.62
ROUTER15	10.138.0.63
ROUTER16	10.138.0.64
ROUTER17	10.138.0.65
ROUTER18	10.138.0.66

ROUTER19	10.138.0.67
ROUTER20	10.138.0.68
ROUTER21	10.138.0.69
ROUTER22	10.138.0.70
ROUTER23	10.138.0.71
ROUTER24	10.138.0.72
PJ6	10.138.0.73
WAP7	10.137.0.74

8. Embedded Room

Devices	IP Address
PC113	10.140.0.1
PC114	10.140.0.2
PC115	10.140.0.3
PC116	10.140.0.4
PC117	10.140.0.5
PC118	10.140.0.6
PC119	10.140.0.7
PC120	10.140.0.8
PC121	10.140.0.9
PC122	10.140.0.10
PC123	10.140.0.11
PC124	10.140.0.12
PC125	10.140.0.13
PC126	10.140.0.14
PC127	10.140.0.15
PC128	10.140.0.16
PC129	10.140.0.17
PC130	10.140.0.18
PC131	10.140.0.19
PC132	10.140.0.20
PC133	10.140.0.21
PC134	10.140.0.22
PC135	10.140.0.23
PC136	10.140.0.24
PC137	10.140.0.25
PC138	10.140.0.26
PC139	10.140.0.27
PC140	10.140.0.28
PC141	10.140.0.29
PC142	10.140.0.30
PJ7	10.140.0.31
WEBCAM	10.140.0.32

SCANNER	10.140.0.33
SMARTLOCK	10.140.0.34
MONITOR2	10.140.0.35
PRINTER2	10.140.0.36
WAP8	10.140.0.37

9. Hybrid Classroom

Devices	IP Address
PJ7	10.142.0.1
MONITOR3	10.142.0.2
MONITOR4	10.142.0.3
PROJECTOR	10.142.0.4
WAP9	10.142.0.5

10. Server Room

Devices	IP Address
SV1	10.128.0.1
SV2	10.128.0.2
SV3	10.128.0.3
SV4	10.128.0.4
SV5	10.128.0.5
SV6	10.128.0.6
SV7	10.128.0.7
SV8	10.128.0.8
SV9	10.128.0.9
SV10	10.128.0.10

5.3 Reflection of Task 5

From this task we learned how to do IP address subnetting with the AND operation, which allow us to identify the network address, broadcast address and the usable IP address. Due to high number of work areas, we borrow 3 extra bits from the network portion of the original IP address so that these derived subnets can accommodate more work areas and ensure seamless connection throughout the faculty building. The subnetted IP address are allocated to each work area and thus usable IP address is given to the respective end devices.

In conclusion we derived a total of 8 subnet address for 10 work areas. These subnets will ensure security, data integrity and ease management of IP addresses allocated to each work areas.

6.4 Conclusion

In all, this project enabled us to determine an optimal floor plan for the four labs, including proper dimensions and appropriate internet connectivity. A short survey was also conducted in an attempt to secure the best price available without compromising on the quality of the required equipment. We were also able to effectively determine subnet addresses without problems.

The strengths of our floor plan are enhanced teamwork through thoughtful table arrangements, wireless scanners, and printers that streamline students' work processes. The addition of touch-screen whiteboards enables interactive learning, while strategically placed tables and wiring enable efficient maintenance and repair. These elements create an environment that is both functional and student friendly.

However, there are a few drawbacks to the design. The hybrid setup lacks key modern resources such as 3D printers, which are important in prototyping and design courses. Secondly, there is a lack of external monitors that would be helpful for multitasking on projects or during coding. Finally, speech-to-text devices that could support accessibility needs with notetaking during lectures were not part of the design.

Despite such challenges, the project was efficiently and effectively completed on schedule with the facilitation of online collaboration tools and the guidance of our lecturer. Constraints developed by the global pandemic, including an inability to conduct physical meetings, were well mitigated through teamwork and adaptability.

6.5 Team Members and Responsibilities

Group Name: Router Ranger

Section: 02			
Group Members			
No.	Member Photo	Information	Responsibility
1.	 Choh Jing Yi	Name: Choh Jing Yi Matric No: A23CS0296 Position: Leader	I discussed the requirements for every task with my group members and started Google Meet discussions to make sure everyone understood the objectives. I made sure that everyone was thoroughly understood the project guidelines and assessment rubric throughout these meetings. I distributed tasks equally among the members and setting clear due dates to maintain progress. I continuously promoted efficient communication and teamwork by conducting meetings and progress reports to assist the team. Lastly, I took responsibility for examining every deliverable to ensure the project was completed on time and met the established goals and standards.
2.	 Neo Li Xin	Name: Neo Li Xin Matric No: A23CS0253 Position: Leader Assistant	As the assistant of our leader Choh Jing Yi, I am responsible for confronting details of each task and report with him before final task distribution to the group members. I sometimes prepared the google meeting for group discussion for each task while explain the task details to ensure we understand the rubric and details of each project. Lastly, I oversee checking the writing format and grammar errors, ensuring a polished report before the final submission.
		Name : Pravinraj A/L Sivabathi Matric No: A23CS0171	As an active member, I contributed to key tasks throughout the project. In Task 1, I designed the second-floor layout with spaces like the Embedded Lab, Hybrid Room, Server Room, and Electrical Room, ensuring functionality and efficiency. In Task 2, I developed key questions that addressed wireless network coverage, wired

	Pravinraj A/L Sivabathi		and wireless connectivity, and technical feasibility to guide effective planning and decision-making. In Task 3, I analyzed LAN devices by explaining the roles of routers and switches in providing reliable performance in a network. Then, in Task 4, I identified the connections, cable types, and distribution for each floor and room. Lastly, in Task 5, I planned network distribution to workstations, thus enhancing connectivity across all areas.
	 Ahmad Ziyaad bin Mohd Abbas	Name: Ahmad Ziyaad bin Mohd Abbas Matric No: A23CS0206	As an active member of this group, I was responsible in designing the Hybrid Classroom and ensuring that all the equipment and furnitures followed the specification given. I also was in charge in selecting the equipment need for building our internet connection for this project. I am also involved in the documentation of this project.

6.6 References

Task 2.1

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Task 3.1

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6.7 Appendices

Financial Budget

Devices	Price per unit (RM)	Quantity	Total price (RM)
Router  Figure 6.7.1	799.00	29	23171
Switches  Figure 6.7.2	1445.00	24	34680
Patch panels  Figure 6.7.3	178.88	6	1073.28
Ethernet cables  Figure 6.7.4	4.925 per meter	250	1231.25
Wireless access point	1758.00	10	17580

 Figure 6.7.5			
Fiber optics  Figure 6.7.6	20.00 per meter	486.5	9730
Total			87465.3
Budget			1000000
Remaining Budget			912534.47
Budget Usage			$(87465.3/1000000)$ $\times 100\% = 8.75\%$

Meeting Minutes

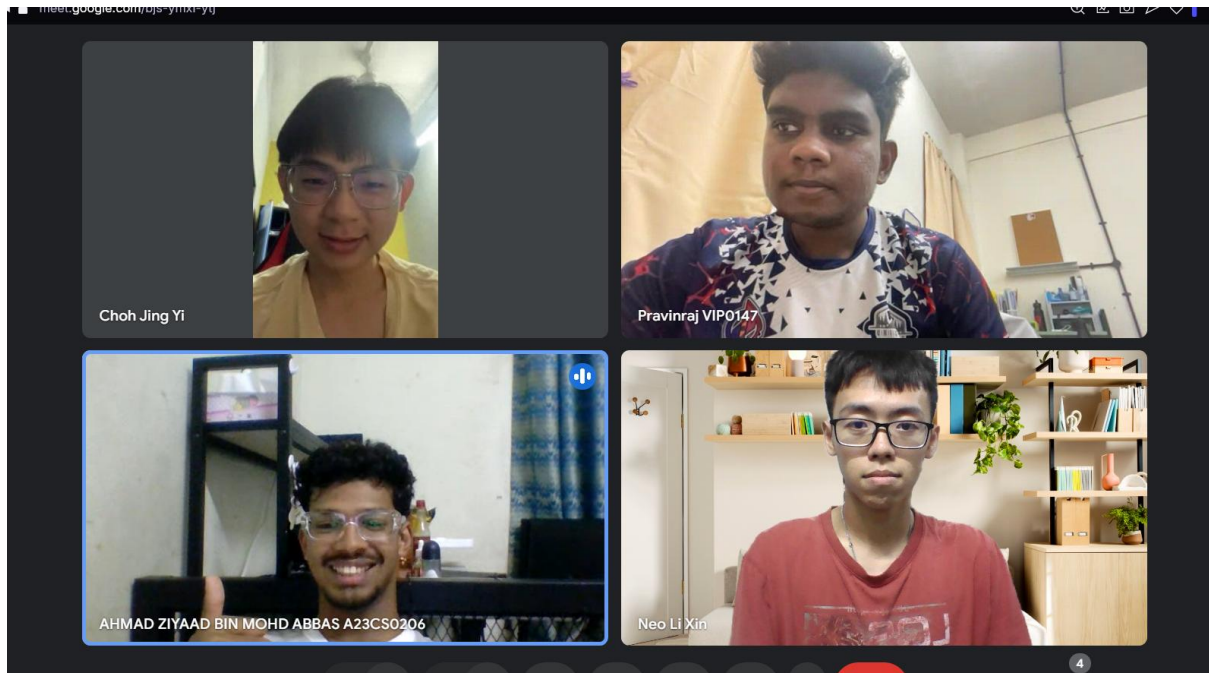
Meeting Minutes Task 1

Meeting Minutes #1

Date/Time	1/11 November 2024 10:00pm
Location Online	Google Meet
Agenda	1. Overviewing task 1 2. Designing the floor plan

		3. Deciding tools for floor plan designing	
		4. Task distribution for each group member	
Attendance			
Name	Time		Reason For Absence
Choh Jing Yi	22 00		-
Neo Li Xin	22 00		-
Pravinraj a/l Sivabathi	22 00		-
Ahmad Ziyaad bin Mohd Abbas	22 00		-
Meeting details			
No.	Discussing item	Item details	Person-In charge
1.	Overviewing the case study of task 1 and make a simple discussion	- Choh Jing Yi show the case study and rubric of the task 1 - Make a simple discussion of the project and make sure everyone group member understands the project requirements and goals	Choh Jing Yi
2.	How to design the floor plan	-All group members find some floor plan template on online as inspiration for design -We also discuss about the design of the network infrastructure and project setup	All group members
3	Discuss what tools for drawing the floor plan	-We use ChatGPT to suggest some tool to designing the floor plan (AutoCAD, SketchUp, FloorPlanner and Visual Paradigm) -We try to use the tools suggested but found it is complicated for drawing and some required subscription payment -Lastly, we select Visual Paradigm as our drawing tool for floor planning -Then we discuss about the shape and place of each lab	All group members
4	Task distribution for each group member	Task: Choh Jing Yi-initial floor plan layout, designing for	All group members

		general lab and student lounge. Oversee progress tracking and timeline management, documenting the meeting minutes and ensure all design logged Neo Li Xin- designing video conference, cisco network lab and stall. Documenting meeting minutes and review the final design layout. Pravinraj a/l Sivabathi- designing embedded lab Ahmad Ziyaad bin Mohd Abbas- designing hybrid classroom	
5	Discussing the place and size of each lab and student lounge	Choh Jing Yi-suggest put general lab at the ground floor for student easier attending classes.	
6	Discussing additional rooms or equipment that can be added to the floor plan.	Choh Jing Yi-suggested adding a toilet for each floor Neo Li Xin-suggest adding some food and drink stall for student and staff Pravinraj a/l Sivabathi-recommended adding a lecturer office for lecturers to rest and complete tasks.	
7	Meeting ended and everyone start to do task distributed	-We ended the meeting at 11.00pm 【 after all the discussion had been done	All group members

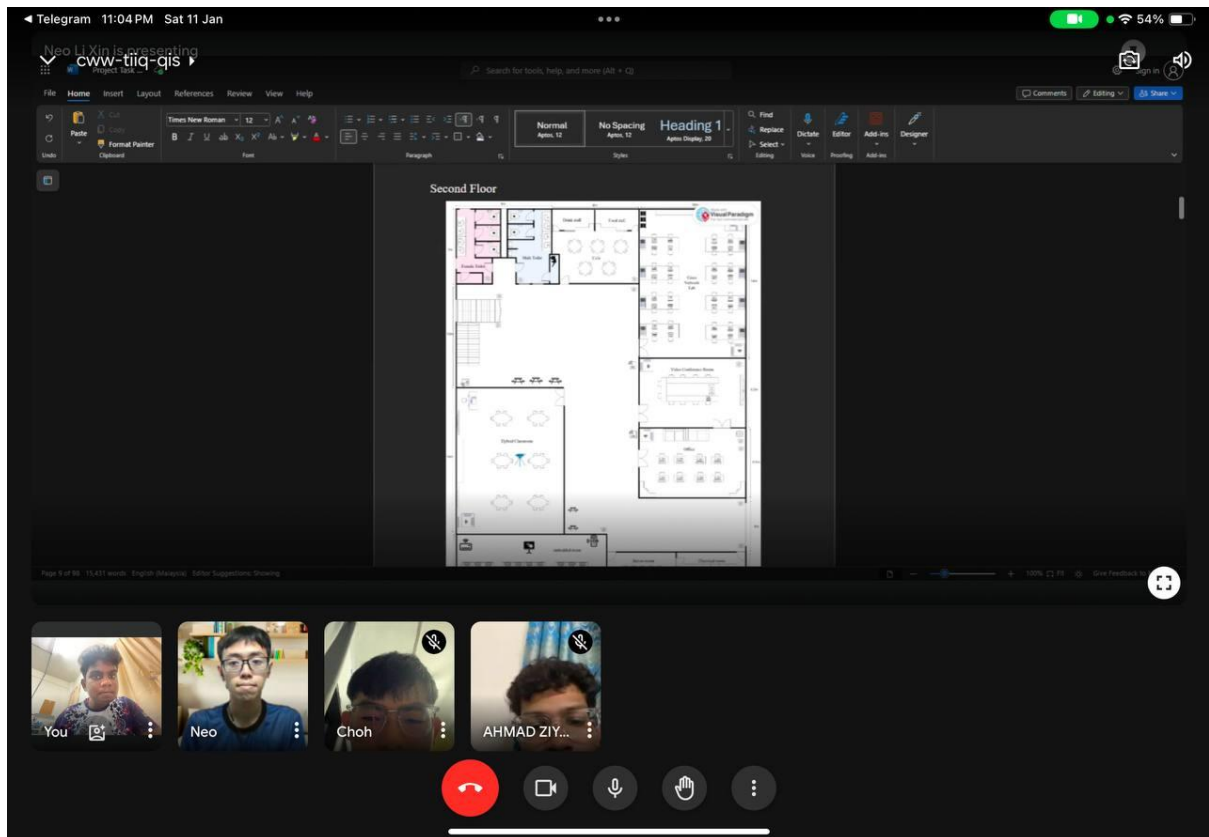


Meeting Photo for Task 1

Meeting Minutes #2

Date/Time		3/11 November 2024 8:00pm	
Location Online		Google Meet	
Agenda		1. Check the progress of each member 2. Edit the design and size of floor planning if found not satisfied 3. Combination of each group member drawing 4. Adding some useful appliances	
Attendance			
Name	Time		Reason For Absence
Choh Jing Yi	20 00		-
Neo Li Xin	20 00		-
Pravinraj a/l Sivabathi	20 00		-
Ahmad Ziyaad bin Mohd Abbas	20 00		-
Meeting details			
No.	Discussing item	Item details	Person-In charge

1.	Check the progress of each group member	- Share the floor planning drawing on visual paradigm	Choh Jing Yi
2.	Edit the design and size of each floor	-Make sure everyone put the scale of distance for each lab and others room -Put some decoration at the corner of the room or lab	All group members
3.	Further improvisation on the floor planning and room equipment	-Choh Jing Yi suggest adding some decoration like flowers and benches -Pravinraj suggest adding electrical and server room -Neo Li Xin suggest the addition of tracking camera that follows the speaker and project it onto the smart TV in the hybrid classroom.	Choh Jing Yi, Pravinraj, Neo Li Xin
4.	Final design review	- Ensure that all layouts are accurately drawn, well-connected, and neatly organized. -Ensure scale are properly documented - Ensure that room positioning and layout are logical	All group members
5.	End of meeting	-We ended the meeting at 20.40 after the building design layout is confirmed.	All group members



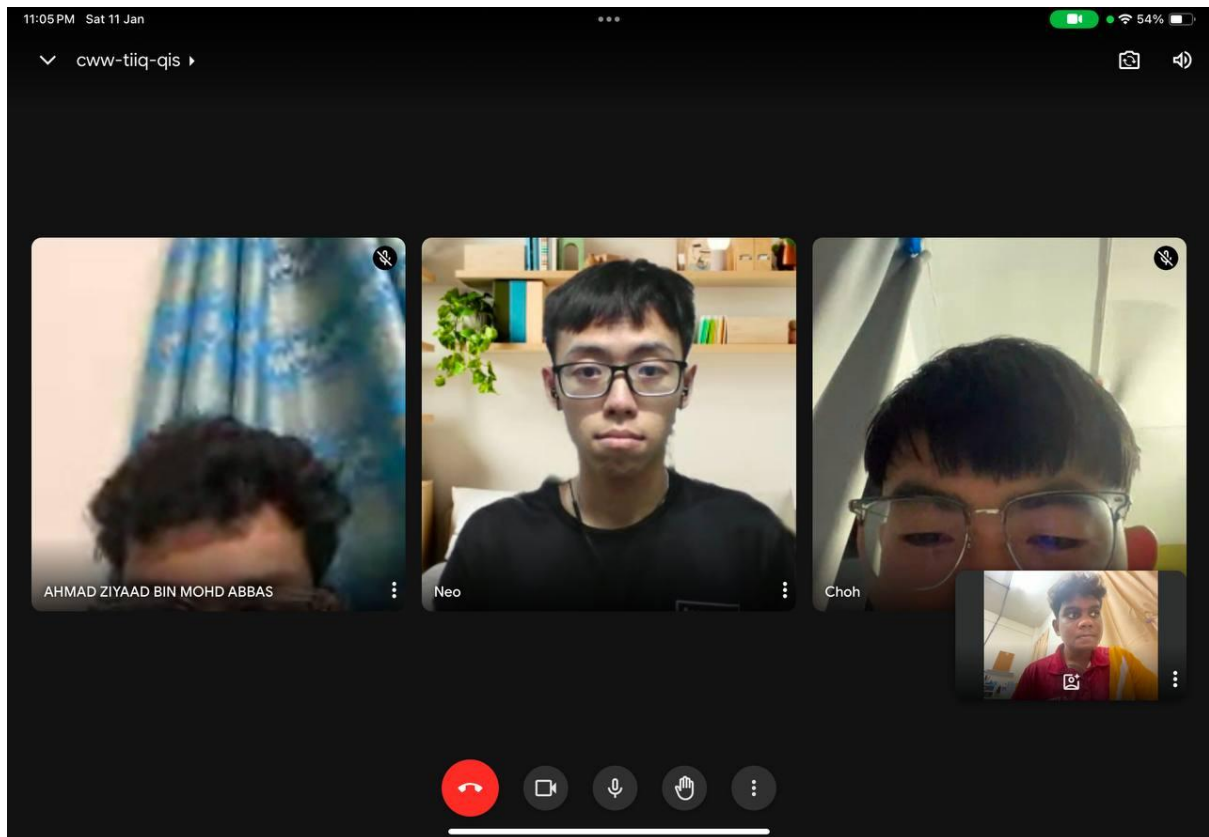
Meeting Photo for Task 1

Meeting Minutes Task 2

Meeting Minutes #1

Date/Time		6/11 November 2024 8:00pm
Location Online		Google Meet
Agenda		1. Overviewing task 2 2. Suggest some question for task 2 3. Task distribution among group members
Attendance		
Name	Time	Reason For Absence
Choh Jing Yi	20 00	-
Neo Li Xin	20 00	-
Pravinraj a/l Sivabathi	20 00	-

Ahmad Ziyaad bin Mohd Abbas		20 00	-
Meeting details			
No.	Discussing item	Item details	Person-In charge
1	Overviewing the case study of task 2 and make a simple discussion	- Choh Jing Yi show the case study and rubric of the task 2 - Make a simple discussion of the project and make sure everyone group member understands the project requirements and goals	Choh Jing Yi
2	Suggestion for questions for task 2	-All members discussed and suggest question using help of google -Every person suggests 3 questions for the task	All group members
3	Choose good question and fulfil the task	-All members vote for desire question and select ten questions out of suggestion	All group members
4	Distribution task	-Choh Jing Yi distributed task equally -Each person does three question and complete one feasibility study	All group members
5	End of meeting	-We ended the meeting at 20.40 after the distribution task confirmed.	All group members

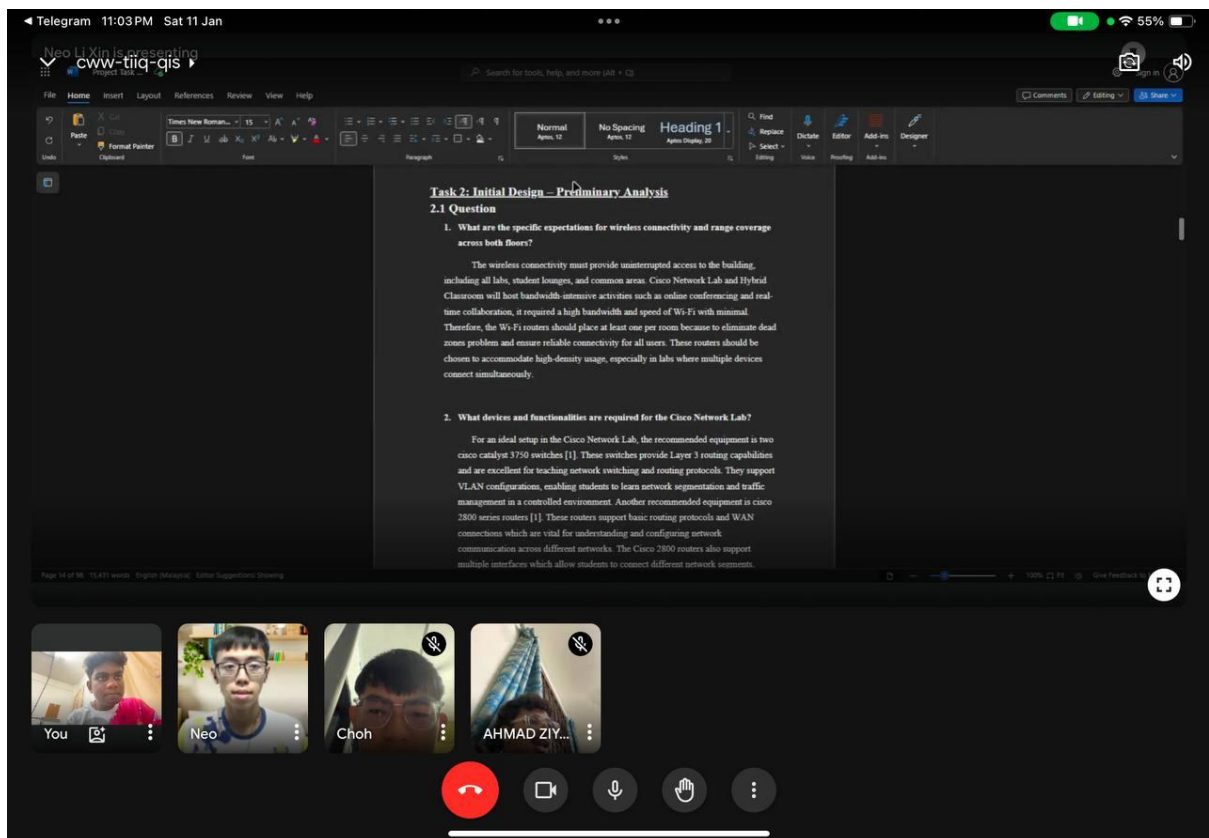


Meeting Photo for Task 2

Meeting Minutes #2

Date/Time		8/11 November 2024 8:00pm	
Location Online		Google Meet	
Agenda		1. Discuss the appropriate answer for each question 2. Discuss the feasibility assessment 3. Write the report for meeting minute	
Attendance			
Name	Time		Reason For Absence
Choh Jing Yi	20 00		-
Neo Li Xin	20 00		-
Pravinraj a/l Sivabathi	20 00		-
Ahmad Ziyaad bin Mohd Abbas	20 00		-
Meeting details			
No.	Discussing item	Item details	Person-In charge

1.	Explain the answer for each question	-Choh Jing Yi review the answer and find the best answer for the question	Choh Jing Yi
2.	Discuss the feasibility assessment	-Choh Jing Yi explain the feasibility assessment from each aspect and make sure the content is fullfill the aspect	Choh Jing Yi
3.	Write the report for meeting minute	-Choh Jing Yi write the report for the meeting minutes - Neo Li Xin added an index	Choh Jing Yi Neo Li Xin
4.	Meeting ended	-We ended the meeting at 21.30 after the review all the report and make a submission	All members



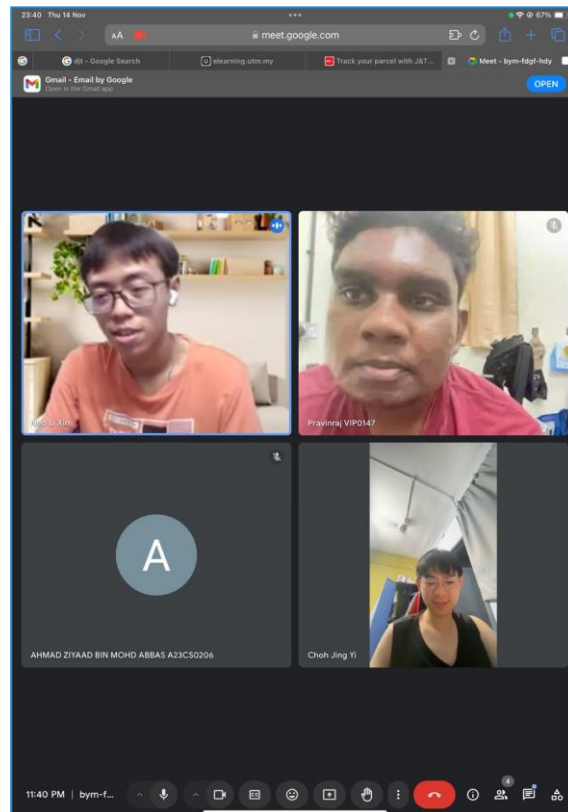
Meeting Photo for Task 2

Meeting Minutes Task 3

Meeting Minutes #1

Date/Time		14/11 November 2024 11:00pm	
Location Online		Google Meet	
Agenda		5. Overviewing task 3 6. List the network devices we want to chose 7. Distribute task among members	
Meeting MC		Neo Li Xin	
Attendance			
Name		Time	Reason For Absence
Choh Jing Yi		23 00	-
Neo Li Xin		23 00	-
Pravinraj a/l Sivabathi		23 00	-
Ahmad Ziyaad bin Mohd Abbas		23 00	-
Meeting details			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Overviewing task 3 and make a simple discussion	-Neo Li Xin share the task 3 information and rubric -All members read the task clearly and try to understand it	All members
2	Suggest the network devices we can use in the building	-In the task, there have list that routers, switches, patch panels, wireless devices, and cables. So, Choh Jing Yi and Neo Li Xin suggest another network devices we can use based on the task 1 floor planning.	Choh Jing Yi and Neo Li Xin
3	Task distribution for each group members	-Choh Jing Yi distribute task to each member equally. -Choh Jing Yi do routers, switches, patch panels and cables	All members

		-Neo Li Xin do wireless access point and fiber optics.	
4	Meeting ended	We end our meeting at 11.30pm after all the discussion has been done.	All members



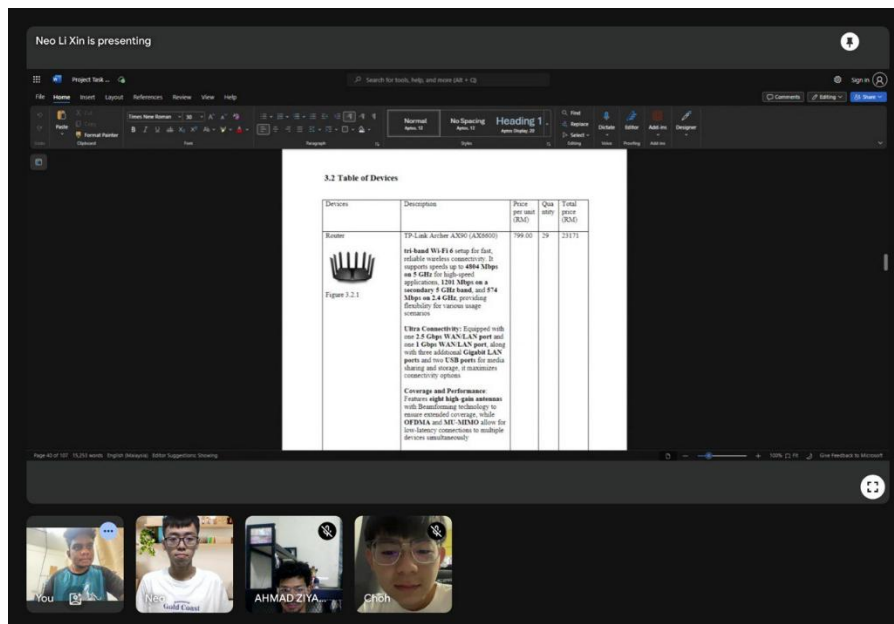
Meeting Photo for Task 3

Meeting Minutes #2

Date/Time	19/11 November 2024 10:00pm
Location Online	Google Meet
Agenda	1. Review the current progress of each member. 2. Check if the suggested product for comparison is suitable.

		3. Decide the quantity and calculate the total price of each device. 4. Check the reference format to see if it matches.	
Meeting MC		Choh Jing Yi	
Attendance			
Name	Time		Reason For Absence
Choh Jing Yi	22 00		-
Neo Li Xin	22 00		-
Pravinraj a/l Sivabathi	22 00		-
Ahmad Ziyaad bin Mohd Abbas	22 00		-
Meeting details			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Review the current progress of each group member	Choh Jing Yi share the part we do together in the word document	All members
2	Check and suggest the network devices that is suitable as an example	-All members review the network devices that proposed - Choh Jing Yi evaluate the features of the selected devices to ensure they are cost-effective and have the durability to last over the long term. -Neo Li Xin suggest the brand of network devices should be reliable and long warranty -Choh Jing Yi suggest the quantity of each network devices based on the requirement -Choh Jing Yi check the comparison of the product suggested is considerable	All members
3	Calculate the total price of each network devices	-Neo Li Xin calculate the total price and make sure it is lower budget and cost effective	Choh Jing Yi,Neo Li Xin

		-Choh Jing Yi leads a discussion to finalize the decision on the most suitable network devices.	
4	Check the reference format to see if it matches.	- Neo Li Xin check the references and make sure each reference is labelled with a number and arranged in order	Neo Li Xin
5	Meeting ended	We end our meeting at 11.30pm after all the discussion has been done.	All members



Meeting Photo for Task 3

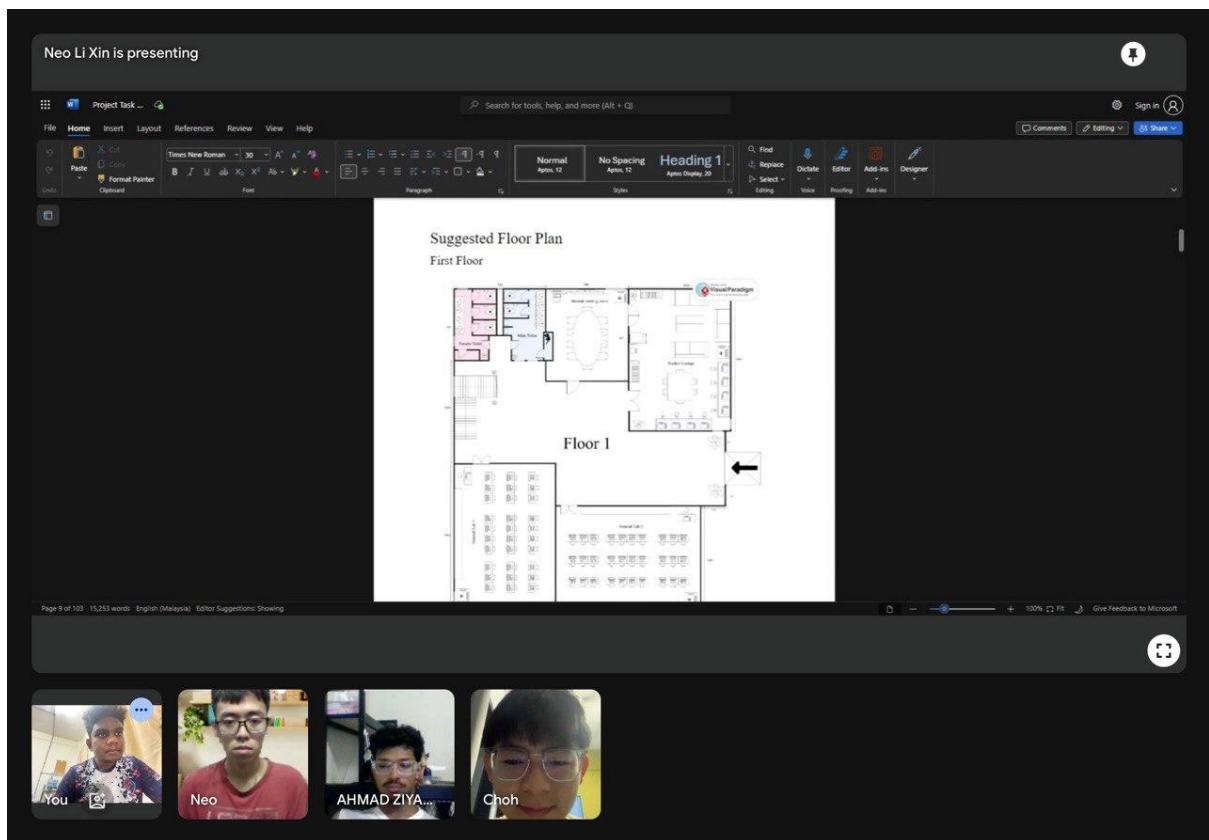
Meeting Minutes Task 4

Meeting Minutes #1

Date/Time	14/12 December 2024 10:00pm
Location Online	Google Meet

Agenda		1. Overviewing task 4 2. Task distribution	
Meeting MC		Choh Jing Yi	
Attendance			
Name	Time		Reason For Absence
Choh Jing Yi	22 00		-
Neo Li Xin	22 00		-
Pravinraj a/l Sivabathi	22 00		-
Ahmad Ziyaad bin Mohd Abbas	22 00		-
Meeting details			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Overviewing task 4 and make a simple discussion	-Choh Jing Yi share the task 4 information and rubric -All members read the task clearly and understand it	All members
2	Suggest what to include in the diagram	-For the network distribution diagram and floor diagram, we discussed about the drawings criteria and how each network component should connect with each other	Choh Jing Yi and Neo Li Xin
3	Task distribution for each group members	-Choh Jing Yi distribute task to each member equally. -Choh Jing Yi is tasked to create floor diagrams and room level network distribution diagram -Neo Li Xin is tasked to create network distribution diagram and overall floor diagram -Pravin is tasked to identify number of components in each room	All members
4	Meeting ended	We end our meeting at 10.30pm after all the discussion has been done.	All members

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Meeting Photo for Task 4

Meeting Minutes Task 5

Meeting Minutes #1

Date/Time	24 December 2024 10:00pm
Location Online	Google Meet
Agenda	1. Overviewing task 2. Task distribution

		3. Completing Task 5	
Meeting MC		Choh Jing Yi	
Attendance			
Name	Time	Reason For Absence	
Choh Jing Yi	22 00	-	
Neo Li Xin	22 00	-	
Pravinraj a/l Sivabathi	22 00	-	
Ahmad Ziyaad bin Mohd Abbas	22 00	-	
Meeting details			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discussion for Task 5	- Choh Jing Yi shared the information about task 5 and its rubric. -All members read and investigate the task carefully. -Solved the problem raised by group members together through discussion.	All members
2	Suggestion for subnetwork division	- All members suggest their prefer subnetwork division - All members discuss and decide the one that best suits our design.	All members
3	IP Addressing Scheme overview	- All members examine how IP addresses are assigned to each work area.	All members

4	Proceed to the parts of Task 5	-All members work on Microsoft Word together to complete task 5. -At the same time, if any of the group members question and identify any issues or problems then we find ways to encounter problems together.	All members
5	Meeting Ended	-At 12.00pm, the meeting ended after all the discussion and work is done.	All members

Neo Li Xin is presenting

Netcom Project Task 5

Search for tools, help, and more (Alt + Q)

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Comments Catch up Editing Share

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5.2.2 Network distribution for workstation in different work areas.

First Floor:

1. General Lab 1

Devices	IP Address
PC1	10.130.0.1
PC2	10.130.0.2
PC3	10.130.0.3
PC4	10.130.0.4
PC5	10.130.0.5
PC6	10.130.0.6
PC7	10.130.0.7
PC8	10.130.0.8
PC9	10.130.0.9
PC10	10.130.0.10
PC11	10.130.0.11
PC12	10.130.0.12
PC13	10.130.0.13
PC14	10.130.0.14
PC15	10.130.0.15
PC16	10.130.0.16
PC17	10.130.0.17

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Meeting Photo for Task 5

Meeting Minutes Task 6

Meeting Minutes #1

Date/Time		2/1 January 2025 2pm	
Location		N28	
Agenda		1. Overviewing task 6 2. Task distribution	
Meeting MC		Choh Jing Yi	
Attendance			
Name		Time	Reason For Absence
Choh Jing Yi		14 00	-
Neo Li Xin		14 00	-
Pravinraj a/l Sivabathi		14 00	-
Ahmad Ziyaad bin Mohd Abbas		-	No reason
Meeting details			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Overviewing task 6 and make a simple discussion	-Choh Jing Yi share the task 6 information and rubric -All members read the task clearly and understand it	All members
2	Task distribution for each group members	-Choh Jing Yi distribute task to each member equally. -Choh Jing Yi (copy previous task, write introduction) -Neo Li Xin (project background) -Pravin (conclusion)	Choh Jing Yi, Pravin, Neo Li Xin
3	Every group member discusses the previous task we have done and reflect it	-Pravin reflect for task 1 -Choh Jing Yi reflect for task 2 and task 3 -Neo Li Xin reflect for task 4 and 5 -We tell the reflection to each member and write it in report after making sure it	Choh Jing Yi, Pravin, Neo Li Xin

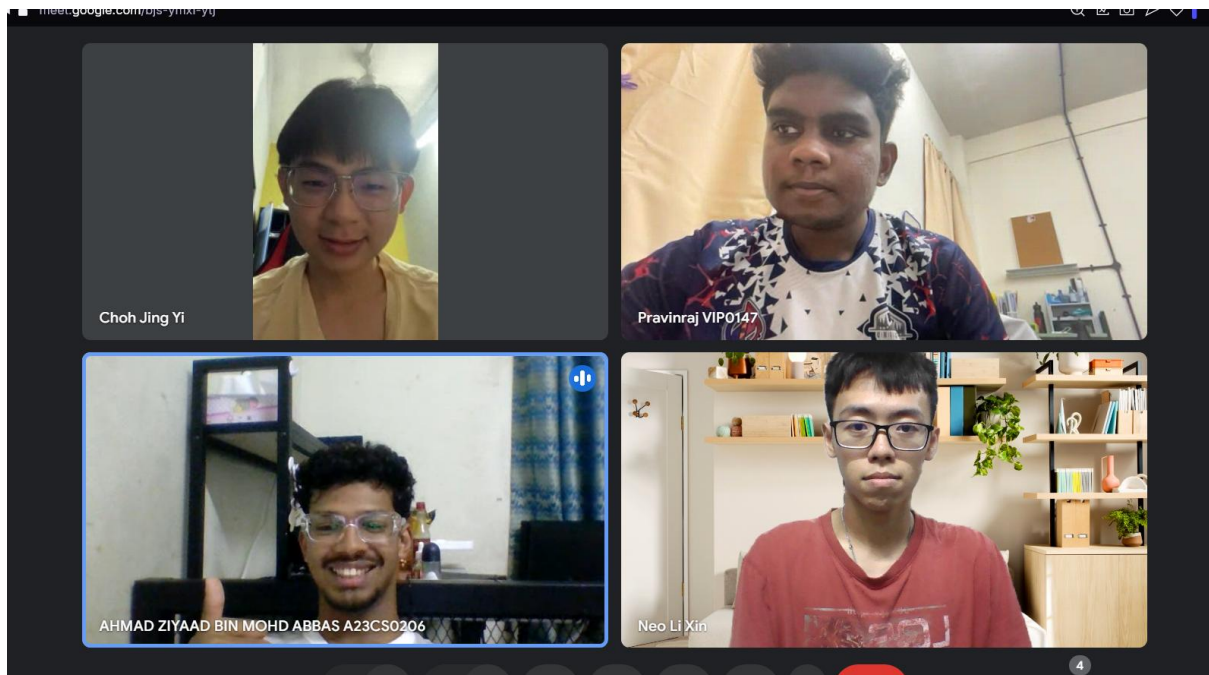
		meets the requirement for the task	
4	Meeting ended	We end our meeting at 3pm after all the discussion has been done.	All members



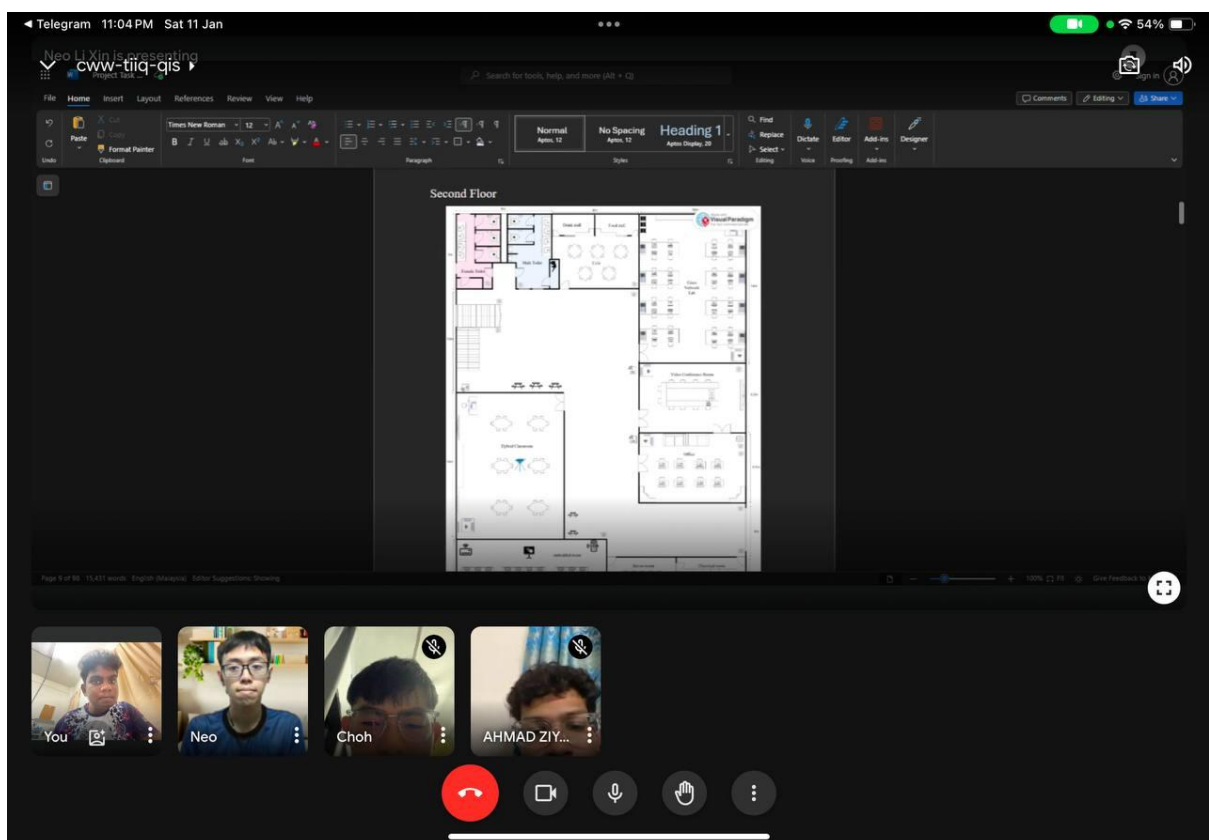
Meeting Photo for Task 6

Picture of Team Working

Task 1



Meeting 1

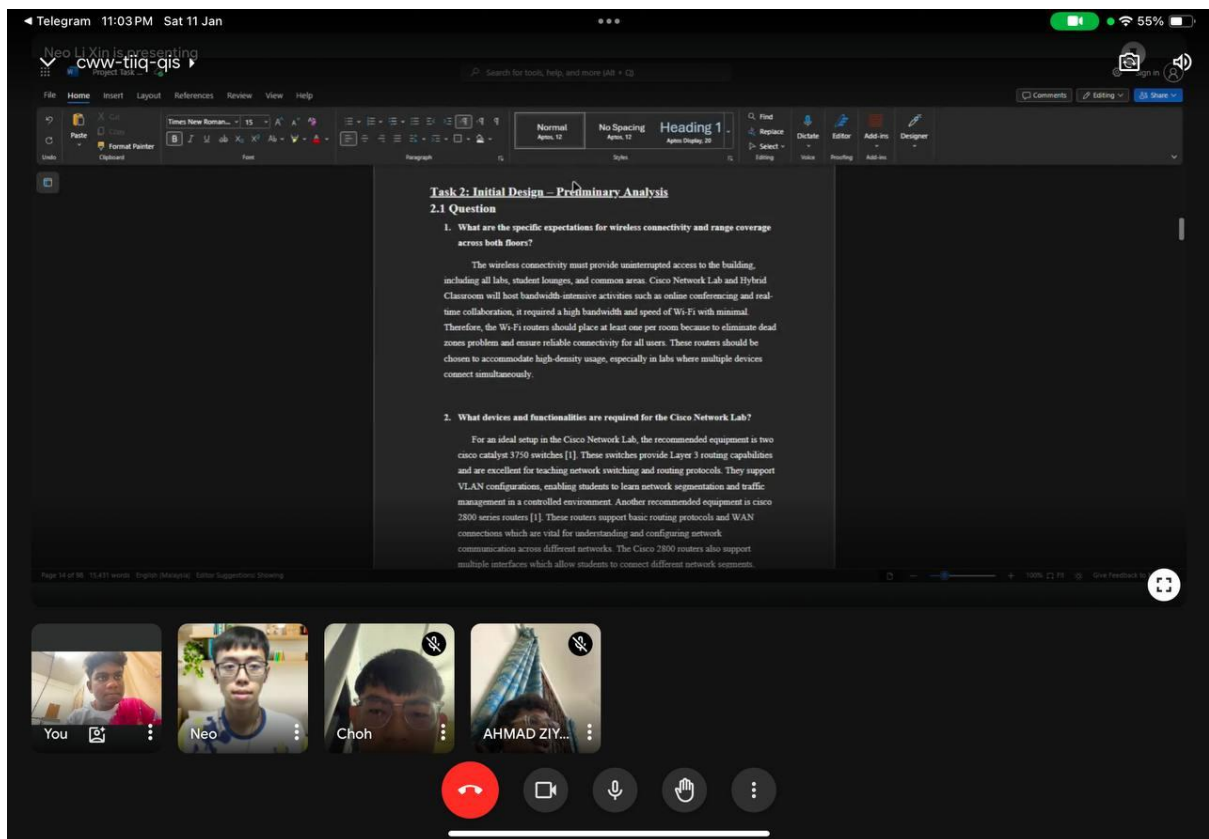


Meeting 2

Task 2

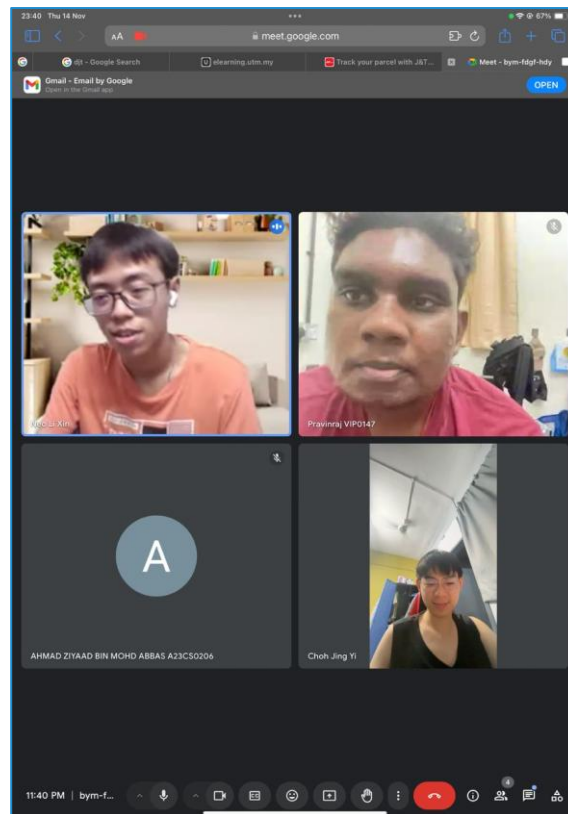


Meeting 1

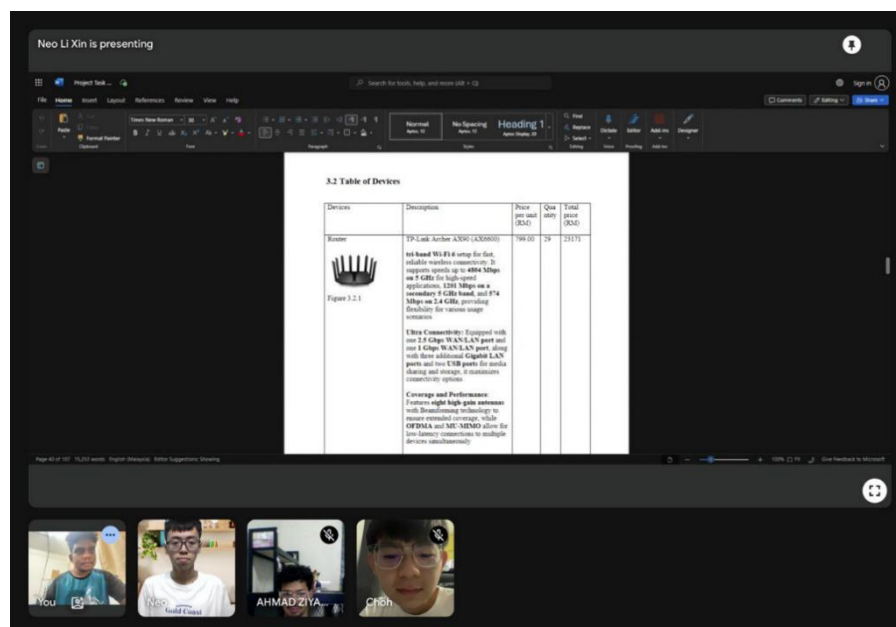


Meeting 2

Task 3

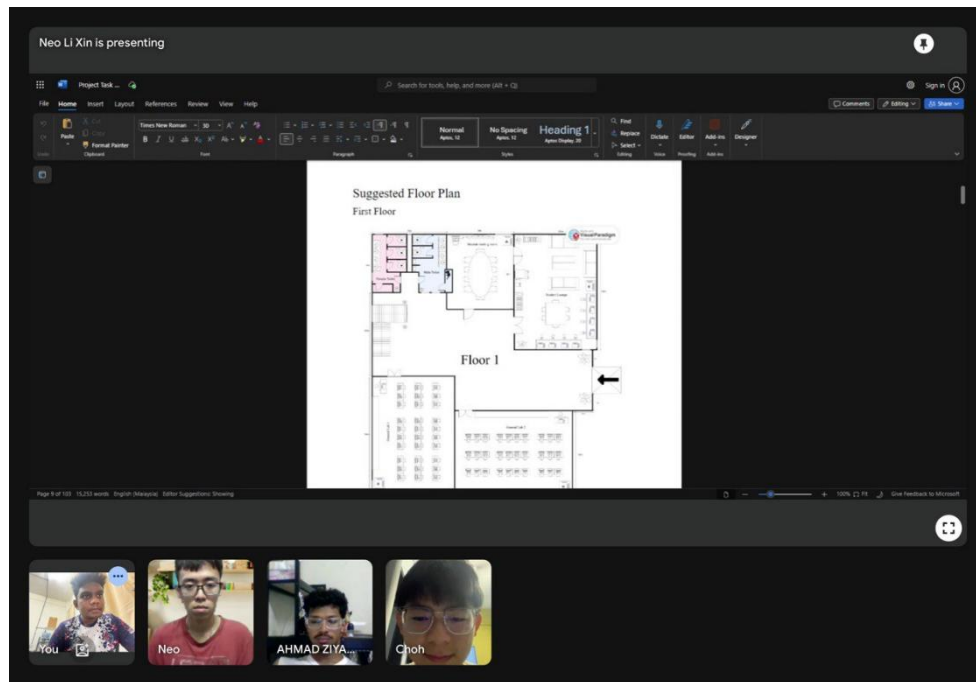


Meeting 1

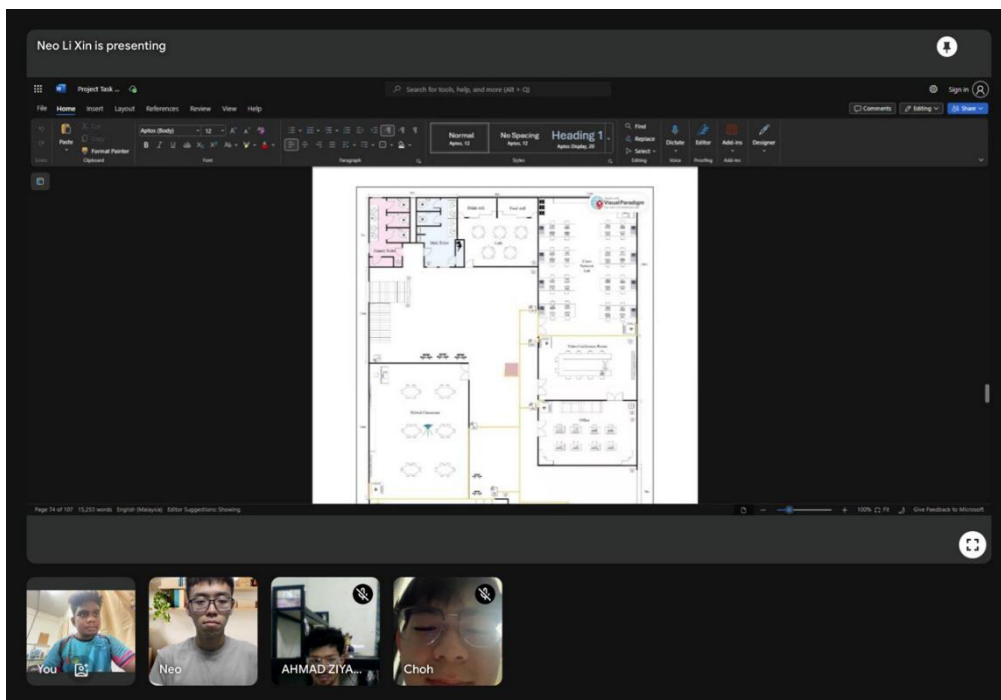


Meeting 2

Task 4

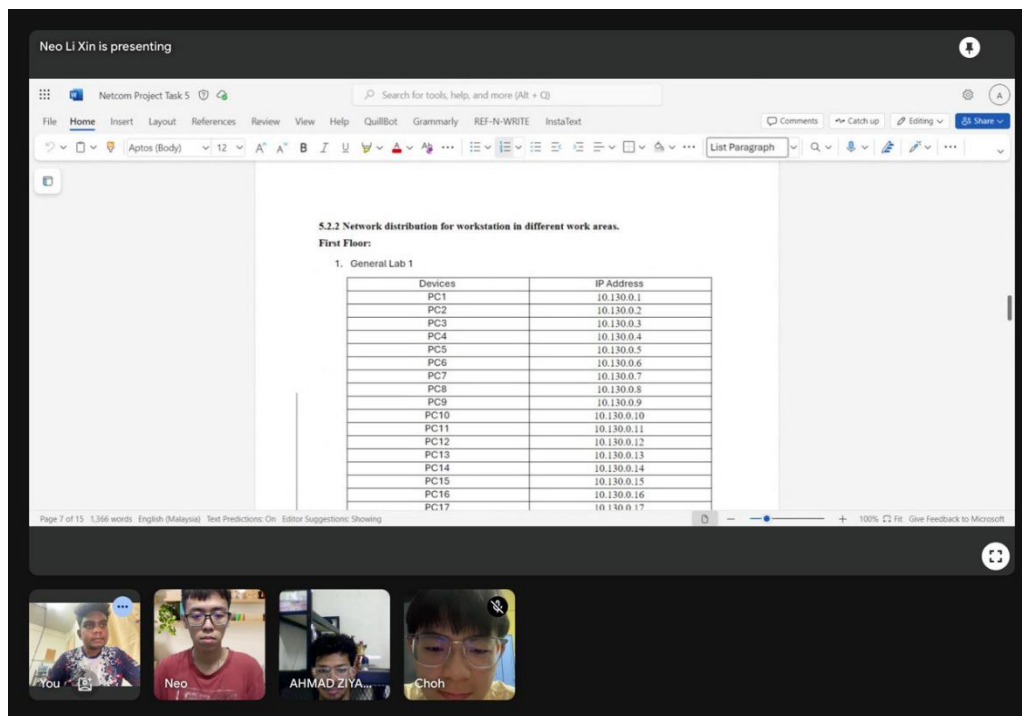


Meeting 1



Meeting 2

Task 5



Meeting 1

Task 6



Meeting 1